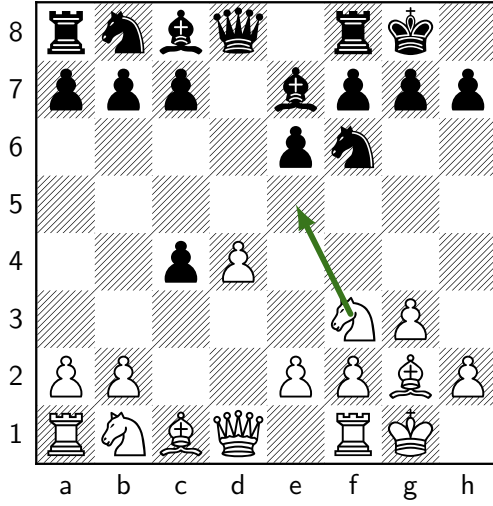


Melanie-XRAY — Catalan

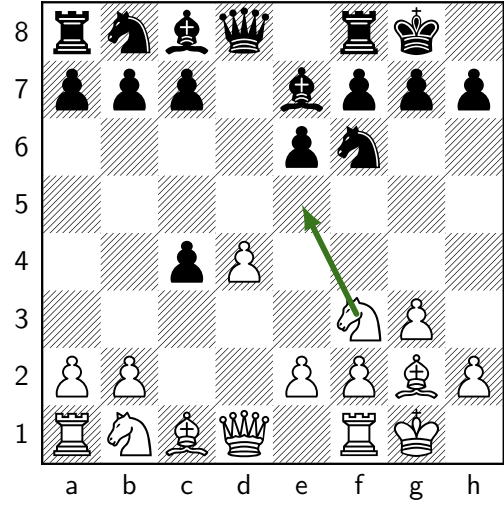
Melanie (Deep_Vision)

Abstract

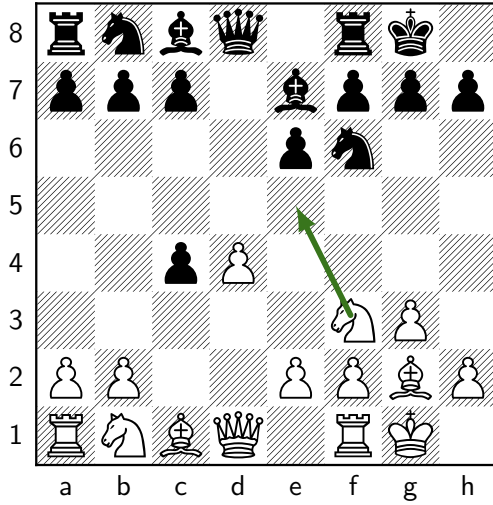
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



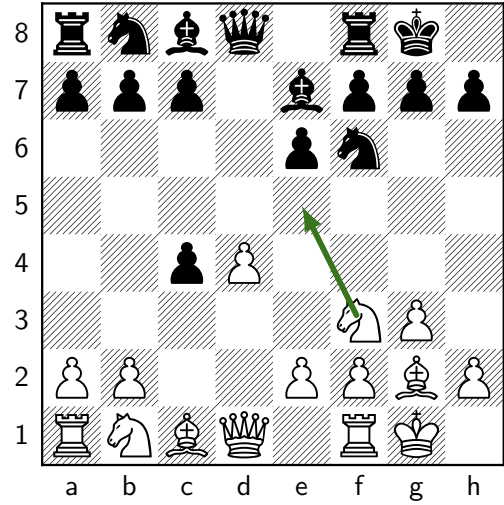
(a)



(b)

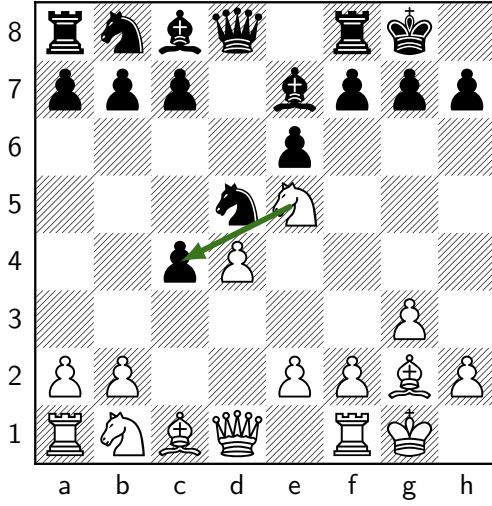


(c)

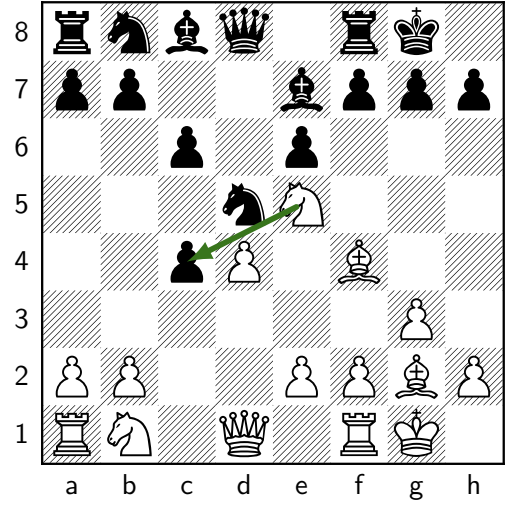


(d)

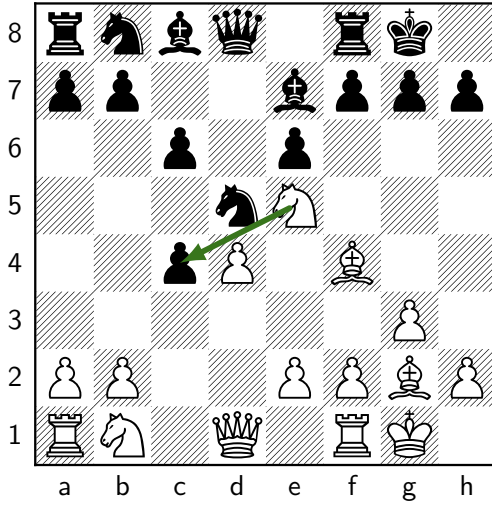
Figure 1: **Move 7, White: Ne5**. (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3464` (ply 12) [structural distance $d=0$, QAP obj=93]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Bc1* as their *Nb1*, our *pc4* as their *Bc1*; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3465` (ply 12) [structural distance $d=0$, QAP obj=93]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Bc1* as their *Nb1*, our *pc4* as their *Bc1*; transferred move f3e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3065` (ply 12) [structural distance $d=0$, QAP obj=93]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Bc1* as their *Nb1*, our *pc4* as their *Bc1*; transferred move f3e5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ne5 (summed QAP objective 1182, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



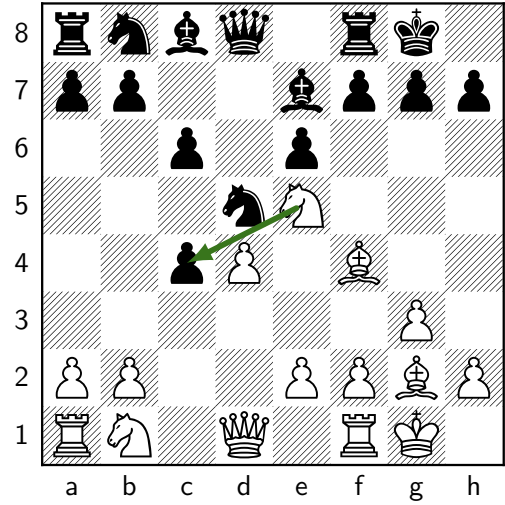
(a)



(b)

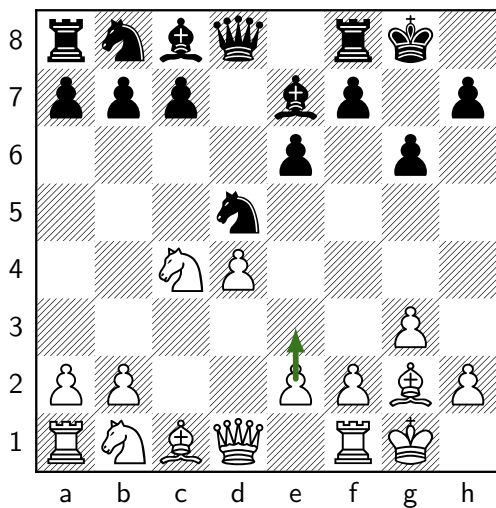


(c)

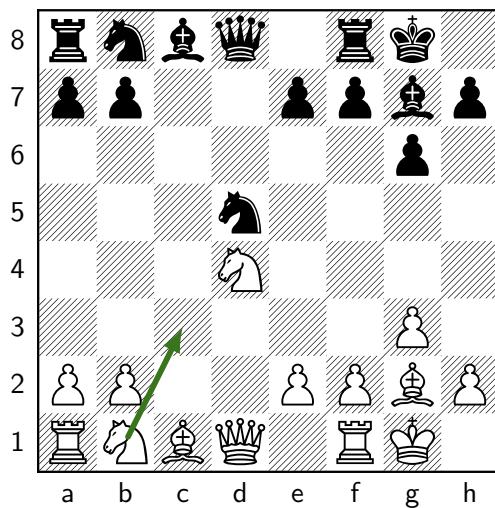


(d)

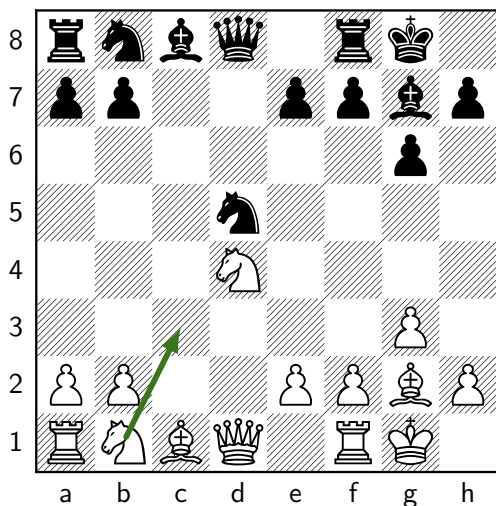
Figure 2: **Move 8, White: Nxc4 (e5c4).** (a) Our position; Melanie plays Nxc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#271` (ply 18) [structural distance $d=33$, QAP obj=53]: the master played Nxc4; under μ our *Ne5* plays the role of that game’s *Ne5*; cross-type roles: our *Bc1* as their *Pd4*, our *pc7* as their *Bf4*; transferred move e5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#272` (ply 18) [structural distance $d=33$, QAP obj=53]: the master played Nxc4; under μ our *Ne5* plays the role of that game’s *Ne5*; cross-type roles: our *Bc1* as their *Pd4*, our *pc7* as their *Bf4*; transferred move e5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3736` (ply 18) [structural distance $d=33$, QAP obj=53]: the master played Nxc4; under μ our *Ne5* plays the role of that game’s *Ne5*; cross-type roles: our *Bc1* as their *Pd4*, our *pc7* as their *Bf4*; transferred move e5c4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nxc4 (summed QAP objective 318, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



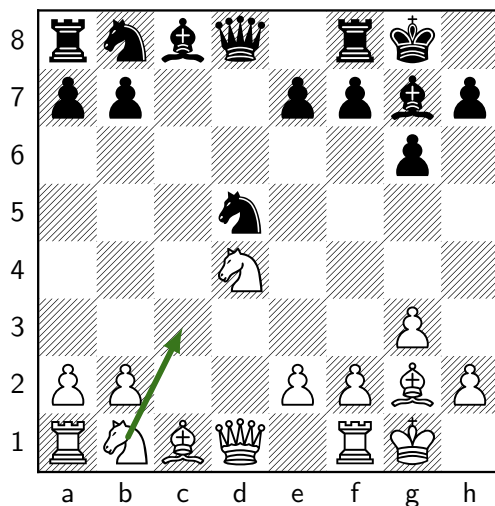
(a)



(b)

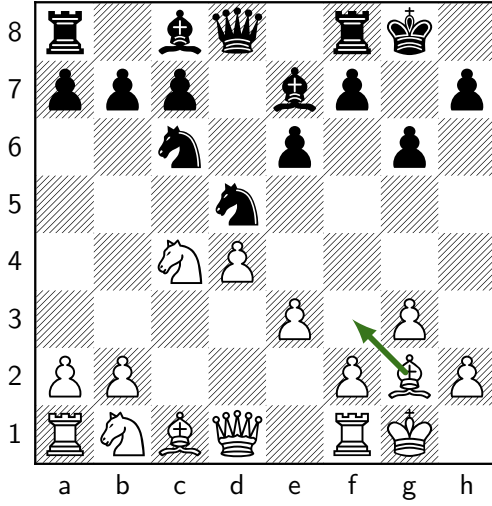


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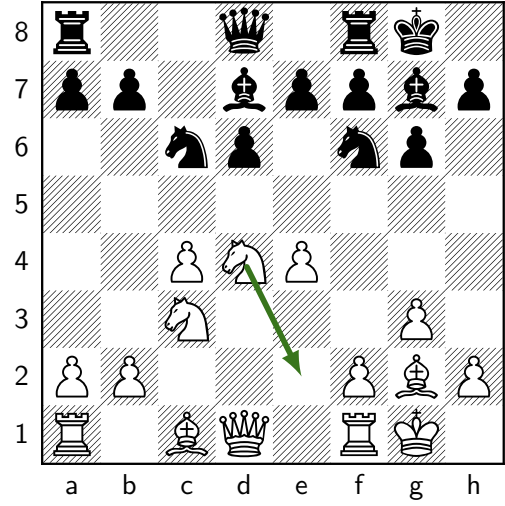


(d)

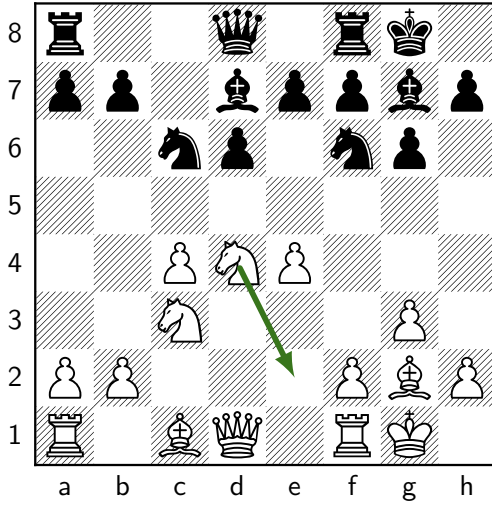
Figure 3: **Move 9, White: e3.** (a) Our position; Melanie plays e3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#874` (ply 16) [structural distance $d=51$, QAP obj=58]: the master played Nc3; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#875` (ply 16) [structural distance $d=51$, QAP obj=58]: the master played Nc3; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move e2e3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2035` (ply 16) [structural distance $d=51$, QAP obj=58]: the master played Nc3; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move e2e3 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to e3 (summed QAP objective 1038, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



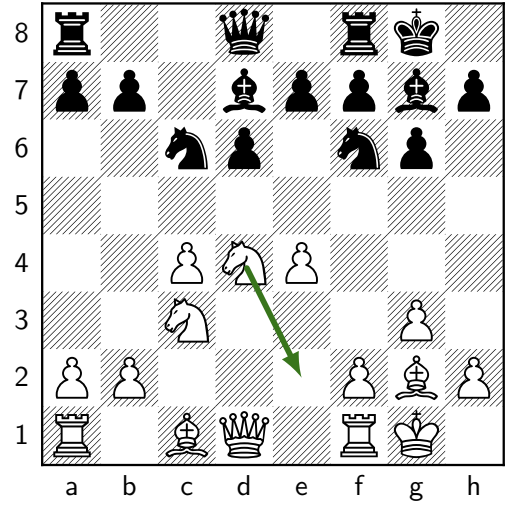
(a)



(b)

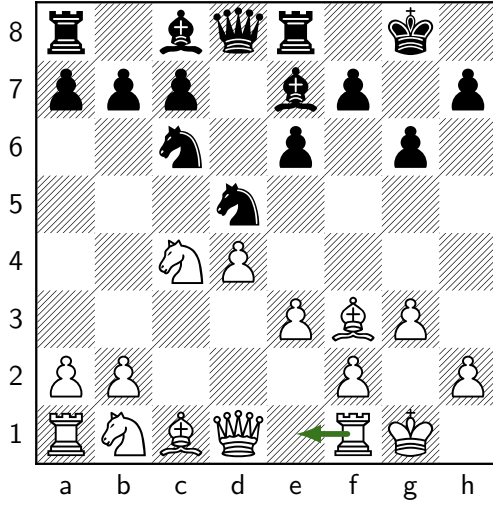


(c)

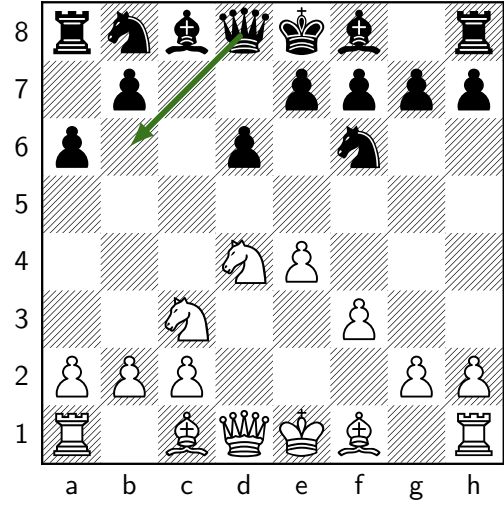


(d)

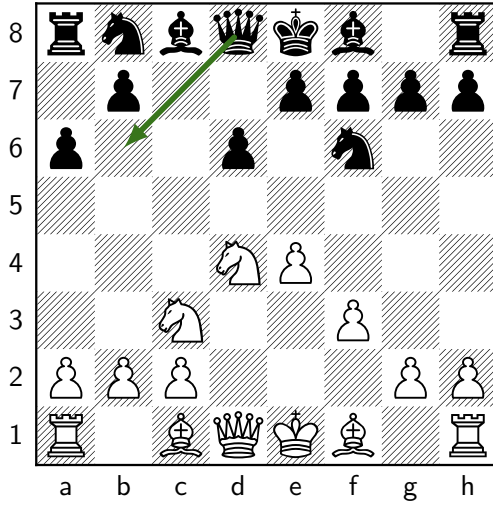
Figure 4: **Move 10, White: Bf3 (g2f3).** (a) Our position; Melanie plays Bf3 (g2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#704` (ply 18) [structural distance $d=55$, QAP obj=30]: the master played Nde2; under μ our Bg2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our pg6 as their Bc1; transferred move g2f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#705` (ply 18) [structural distance $d=55$, QAP obj=30]: the master played Nde2; under μ our Bg2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our pg6 as their Bc1; transferred move g2f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2632` (ply 18) [structural distance $d=55$, QAP obj=30]: the master played Nde2; under μ our Bg2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our pg6 as their Bc1; transferred move g2f3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bf3 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



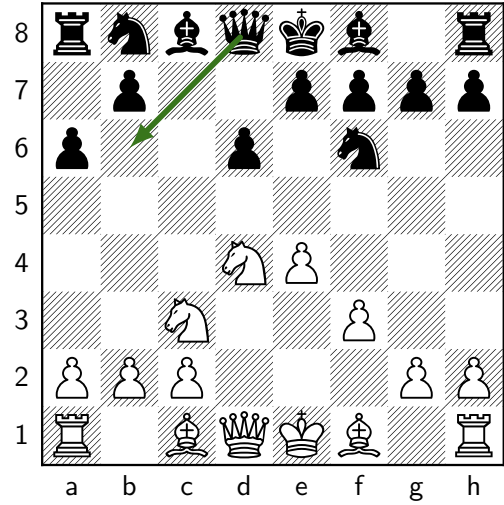
(a)



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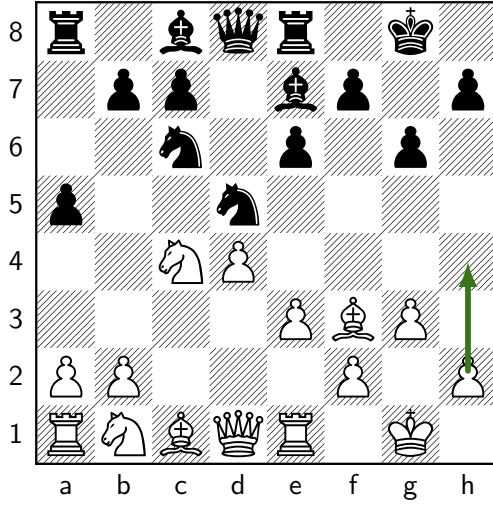


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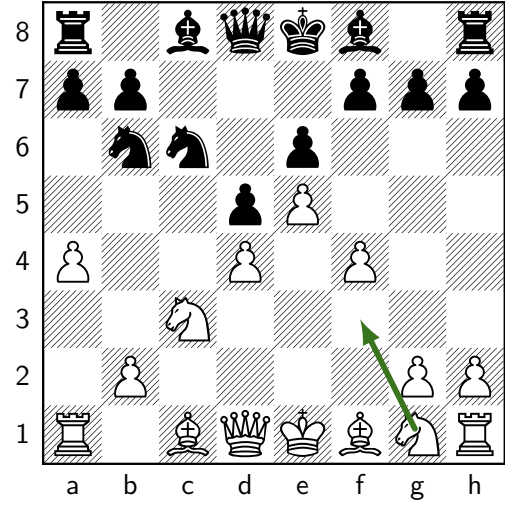


(d)

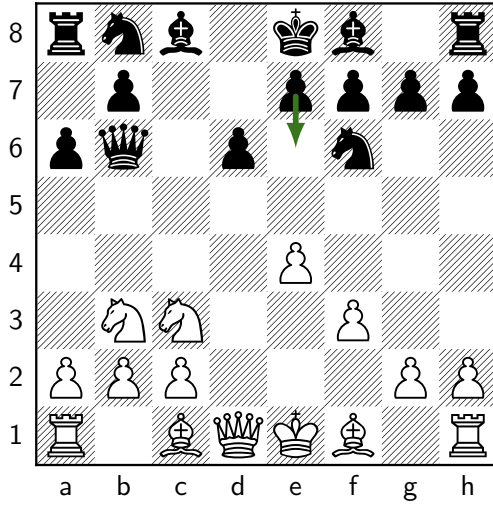
Figure 5: **Move 11, White: Re1.** (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#95 (ply 11) [structural distance $d=69$, QAP obj=15]: the master played Qb6; under μ our Rf1 plays the role of that game's Qd1 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Ra8, our ph7 as their Nb8; transferred move f1e1 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#96 (ply 11) [structural distance $d=69$, QAP obj=15]: the master played Qb6; under μ our Rf1 plays the role of that game's Qd1 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Ra8, our ph7 as their Nb8; transferred move f1e1 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2125 (ply 11) [structural distance $d=69$, QAP obj=15]: the master played Qb6; under μ our Rf1 plays the role of that game's Qd1 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Ra8, our ph7 as their Nb8; transferred move f1e1 (**endorses this move**). Synthesis: 24 retrieved analogs transferred to Re1 (summed QAP objective 368, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



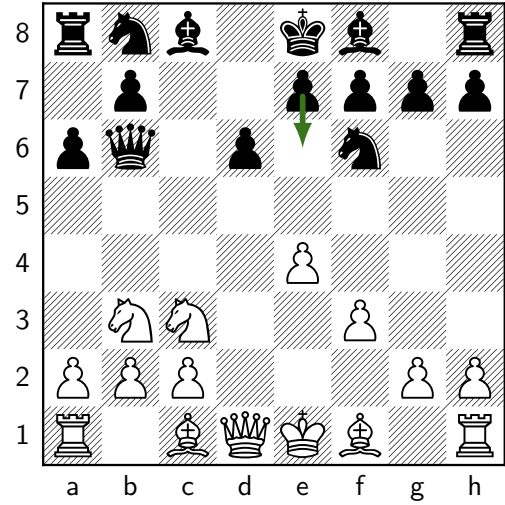
(a)



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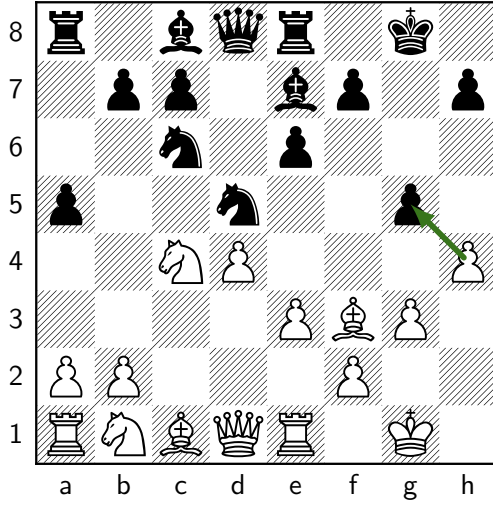


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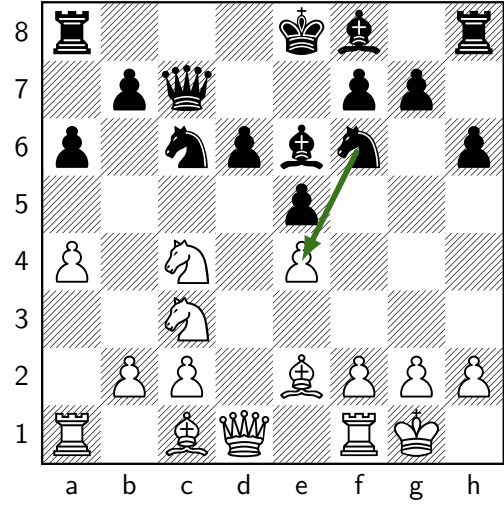


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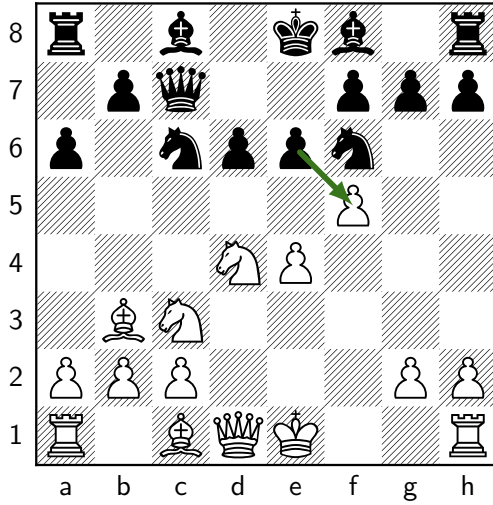
Figure 6: **Move 12, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3805` (ply 18) [structural distance $d=46$, QAP obj=35]: the master played **Nf3**; under μ our *Ph2* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *bc8* as their *Ra1*, our *ra8* as their *Bf1*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#95` (ply 13) [structural distance $d=62$, QAP obj=27]: the master played **e6**; under μ our *Ph2* plays the role of that game’s *Pe2*; cross-type roles: our *Re1* as their *Nb8*, our *Pe3* as their *Bc8*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#96` (ply 13) [structural distance $d=62$, QAP obj=27]: the master played **e6**; under μ our *Ph2* plays the role of that game’s *Pe2*; cross-type roles: our *Re1* as their *Nb8*, our *Pe3* as their *Bc8*; transferred move **h2h4** (**endorses this move**). Synthesis: 13 retrieved analogs transferred to **h4** (summed QAP objective 357, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



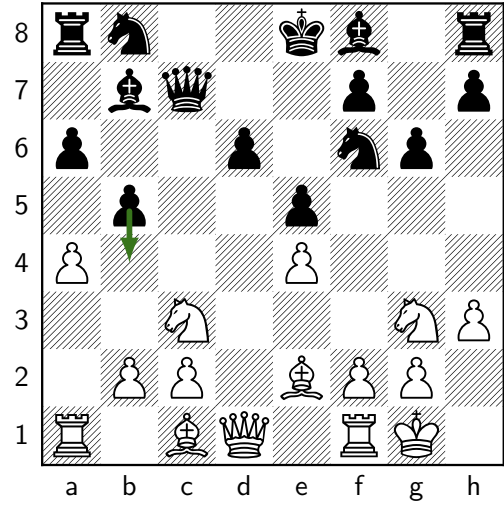
(a)



(b)

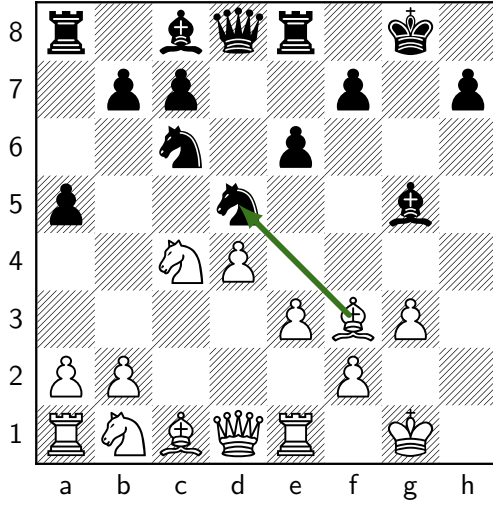


(c)

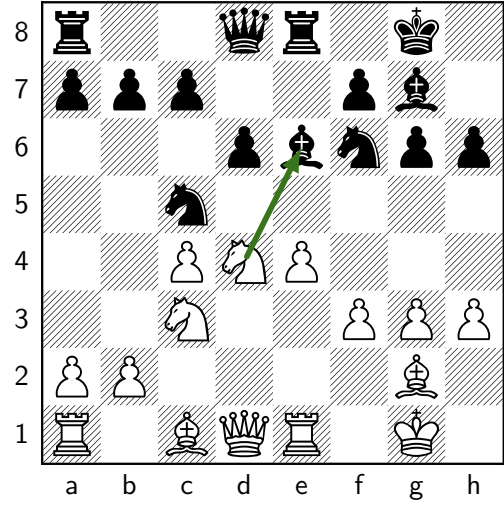


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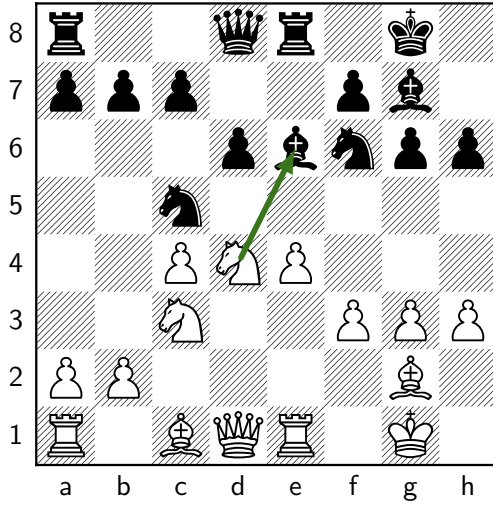
Figure 7: **Move 13, White: hxg5.** (a) Our position; Melanie plays hxg5 (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2458 (ply 21) [structural distance $d=77$, QAP obj=79]: the master played Nxe4; under μ our Ph4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Ra8, our Pd4 as their Bf8; transferred move h4g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#135 (ply 17) [structural distance $d=76$, QAP obj=61]: the master played exf5; under μ our Ph4 plays the role of that game’s Pe3; cross-type roles: our Qd1 as their Ra8, our Bf3 as their Rh8; transferred move h4g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#869 (ply 21) [structural distance $d=74$, QAP obj=53]: the master played b4; under μ our Ph4 plays the role of that game’s Pb4; cross-type roles: our Qd1 as their Ra8, our Re1 as their Nb8; transferred move h4g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to hxg5 (summed QAP objective 350, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



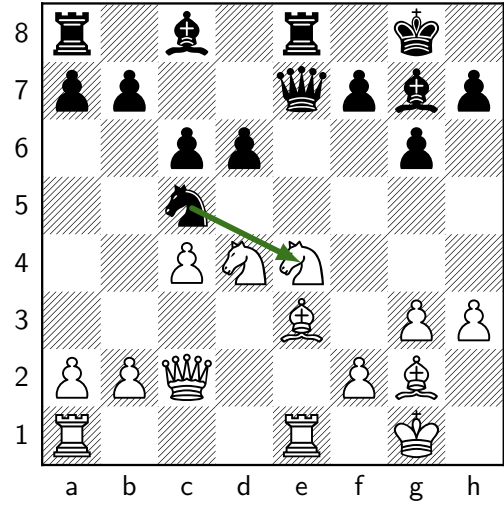
(a)



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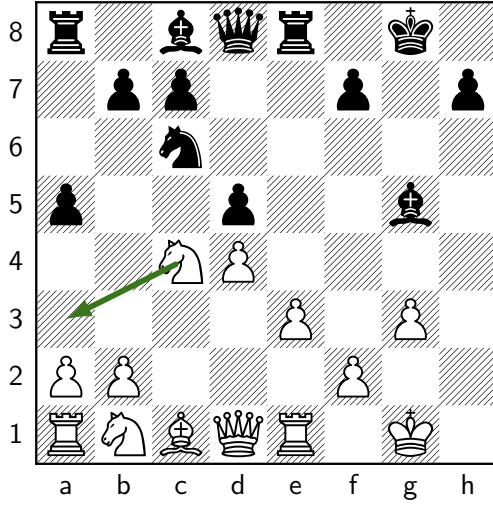


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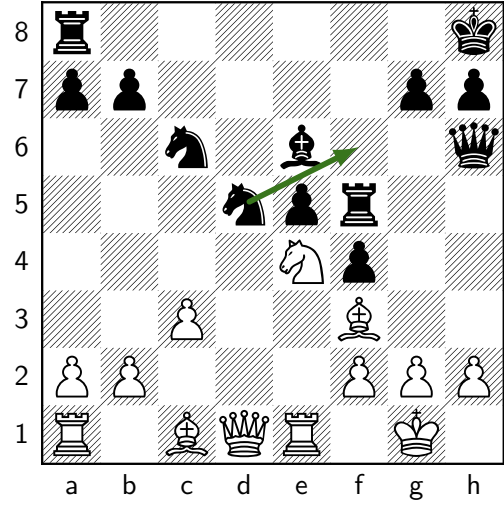


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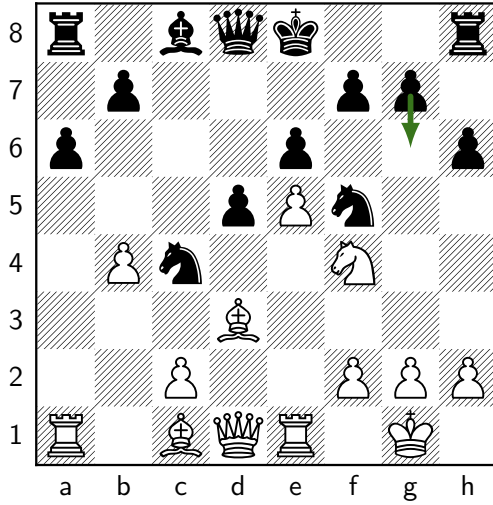
Figure 8: **Move 14, White: Bxd5.** (a) Our position; Melanie plays Bxd5 (f3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3564` (ply 24) [structural distance $d=65$, QAP obj=70]: the master played Nxe6; under μ our Bf3 plays the role of that game's Nd4 (a **cross-type** correspondence); cross-type roles: our pe6 as their Bc1, our Nc4 as their Pa2; transferred move f3d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1044` (ply 24) [structural distance $d=65$, QAP obj=70]: the master played Nxe6; under μ our Bf3 plays the role of that game's Nd4 (a **cross-type** correspondence); cross-type roles: our pe6 as their Bc1, our Nc4 as their Pa2; transferred move f3d5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2043` (ply 27) [structural distance $d=63$, QAP obj=57]: the master played Nxe4; under μ our Bf3 plays the role of that game's Nc4 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Ra8, our ra8 as their Bc8; transferred move f3d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bxd5 (summed QAP objective 217, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



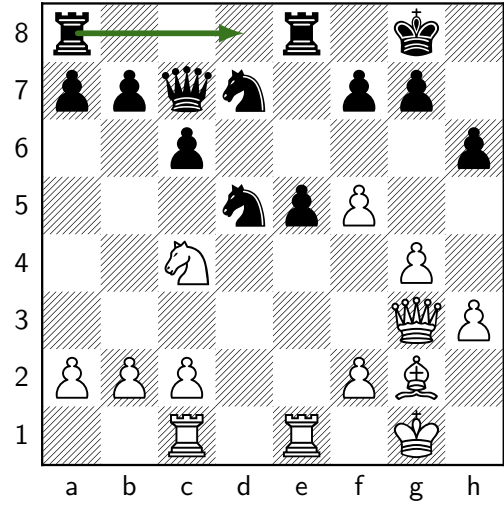
(a)



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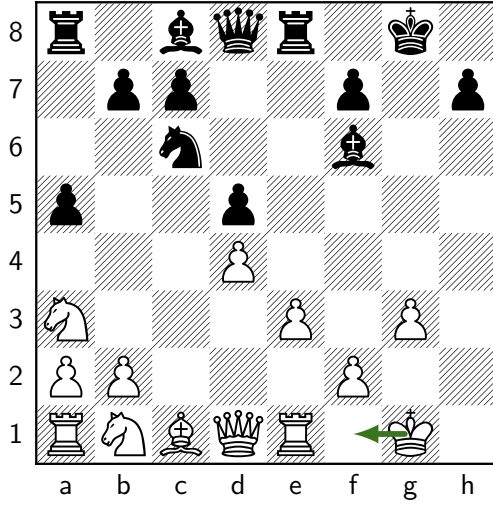


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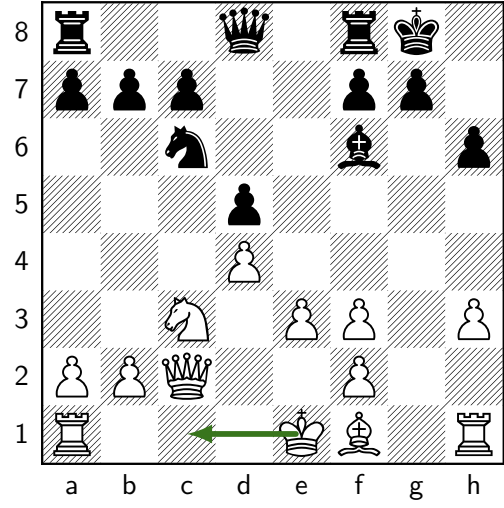


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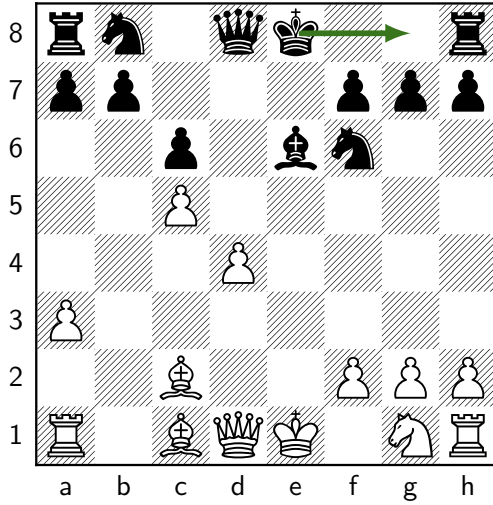
Figure 9: **Move 15, White: Nca3.** (a) Our position; Melanie plays Nca3 (c4a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2211` (ply 43) [structural distance $d=80$, QAP obj=44]: the master played Nf6; under μ our Nc4 plays the role of that game’s Nd4; cross-type roles: our Pg3 as their Ra8, our Ra1 as their Pa7; transferred move c4a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#44` (ply 29) [structural distance $d=80$, QAP obj=61]: the master played g6; under μ our Nb1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Bc8, our Pg3 as their Rh8; transferred move b1a3. (d) Retrieved from `ccrl_top_p0003#332` (ply 37) [structural distance $d=74$, QAP obj=57]: the master played Rad8; under μ our Qd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pa7, our Nb1 as their Pb7; transferred move d1e2. Synthesis: 1 retrieved analog transferred to Nca3 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



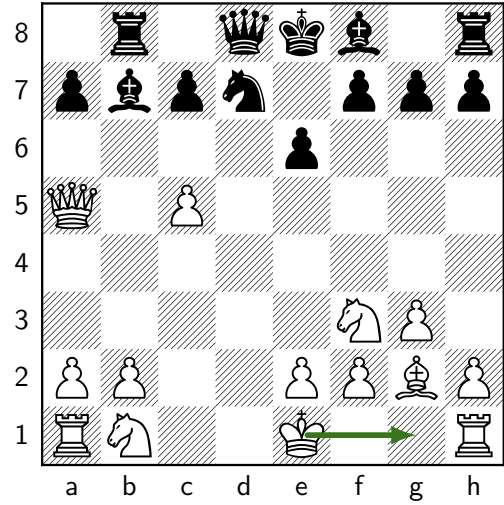
(a)



(b)

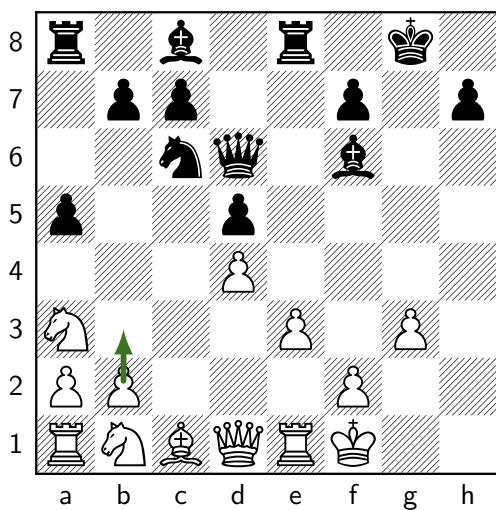


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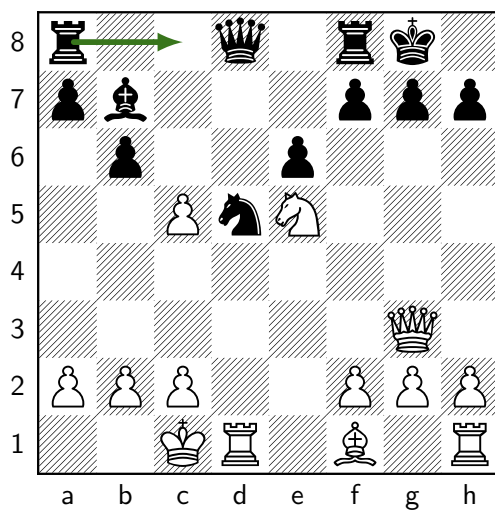


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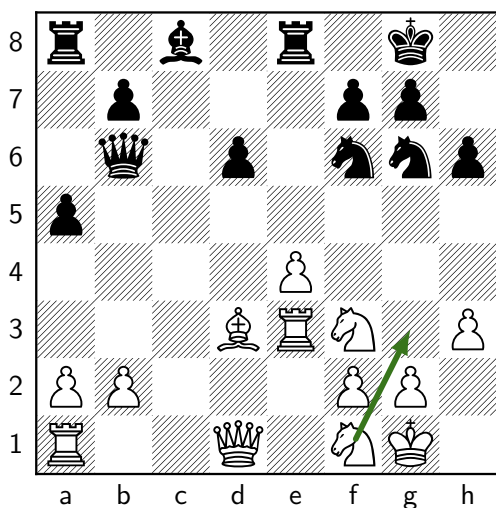
Figure 10: **Move 16, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#420` (ply 22) [structural distance $d=87$, QAP obj=66]: the master played 0-0-0; under μ our *Kg1* plays the role of that game's *Ke1*; cross-type roles: our *Na3* as their *Bf1*, our *Qd1* as their *Rh1*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#318` (ply 21) [structural distance $d=75$, QAP obj=41]: the master played 0-0; under μ our *Kg1* plays the role of that game's *Ke1*; cross-type roles: our *Pf2* as their *Ra8*, our *qd8* as their *Nb8*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#629` (ply 24) [structural distance $d=86$, QAP obj=40]: the master played 0-0; under μ our *Kg1* plays the role of that game's *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Bc1* as their *Nb1*; transferred move g1f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf1 (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



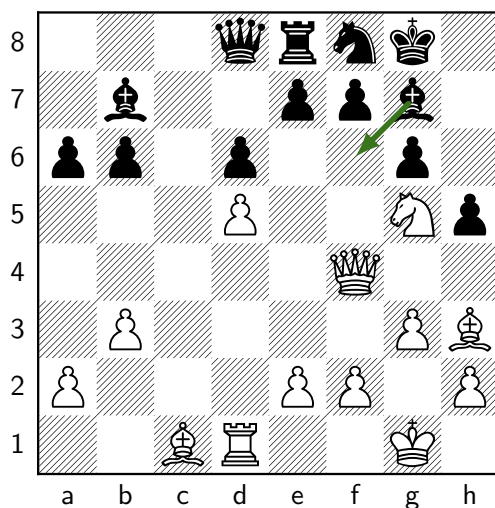
(a)



(b)

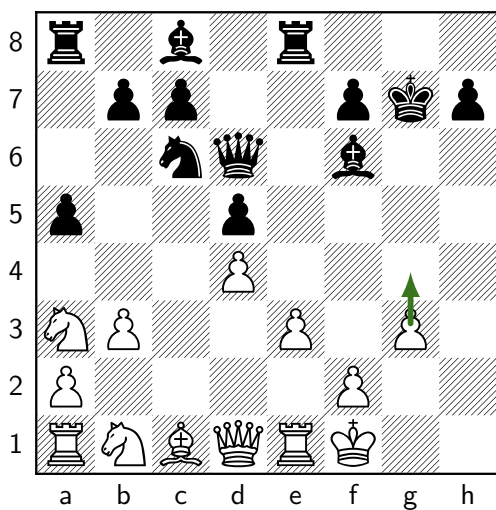


(c)

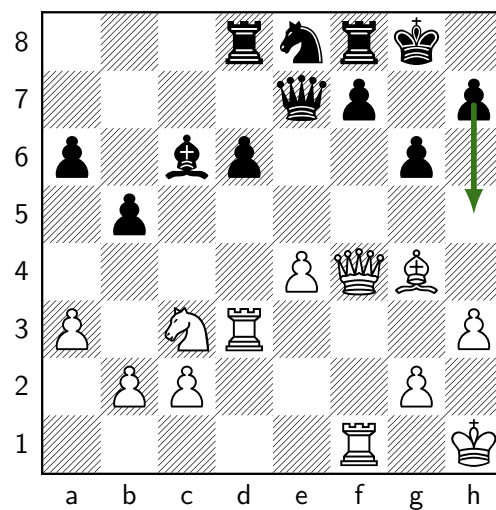


(d)

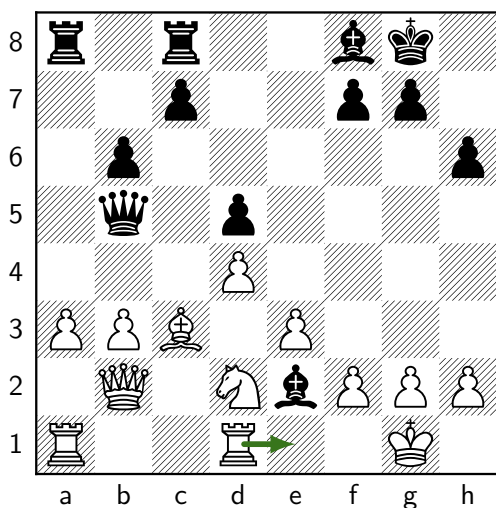
Figure 11: **Move 17, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#517` (ply 27) [structural distance $d=68$, QAP obj=45]: the master played Rc8; under μ our *Pb2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Bb7*, our *qd6* as their *Pb6*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#904` (ply 34) [structural distance $d=67$, QAP obj=35]: the master played Ng3; under μ our *Pb2* plays the role of that game’s *Nf1* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Ra1*, our *Bc1* as their *Pg2*; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1214` (ply 37) [structural distance $d=74$, QAP obj=34]: the master played Bf6; under μ our *Pb2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *ra8* as their *Qd8*, our *Qd1* as their *Re8*; transferred move b2b3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to b3 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



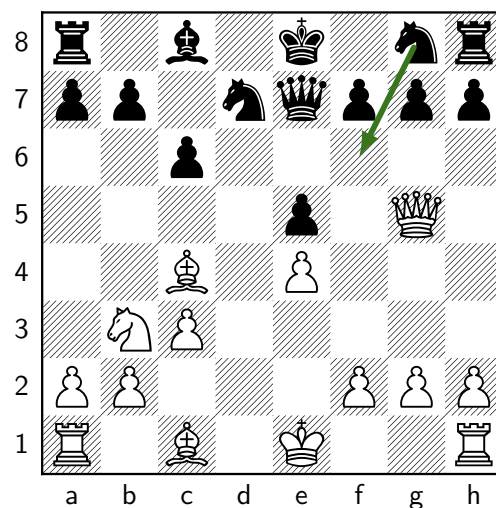
(a)



(b)

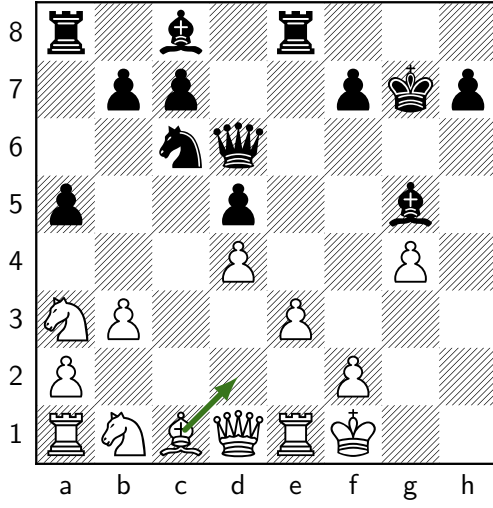


(c)

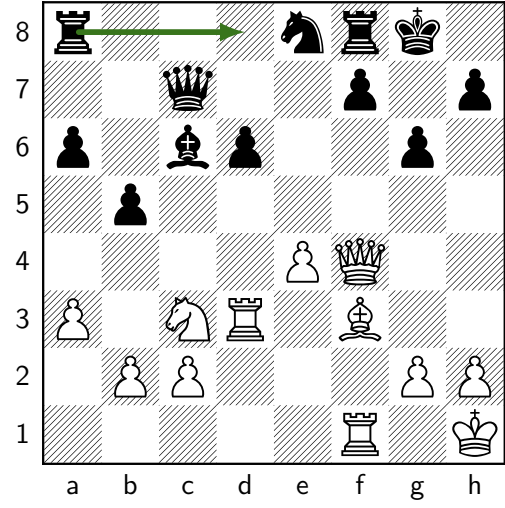


(d)

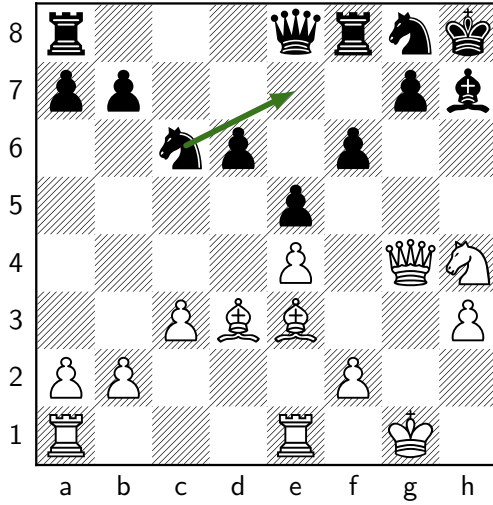
Figure 12: **Move 18, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#451` (ply 43) [structural distance $d=94$, QAP obj=62]: the master played h5; under μ our $Pg3$ plays the role of that game’s $Ph2$; cross-type roles: our $Qd1$ as their $Rd8$, our $Pe3$ as their $Ne8$; transferred move g3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3052` (ply 48) [structural distance $d=89$, QAP obj=36]: the master played Re1; under μ our $Pg3$ plays the role of that game’s $Rd1$ (a **cross-type** correspondence); cross-type roles: our $qd6$ as their $Nd2$, our $pf7$ as their $be2$; transferred move g3g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#990` (ply 21) [structural distance $d=95$, QAP obj=27]: the master played Ngf6; under μ our $Pg3$ plays the role of that game’s $Ng1$ (a **cross-type** correspondence); cross-type roles: our $Pe3$ as their $Bc8$, our $pf7$ as their $Rh8$; transferred move g3g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g4 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



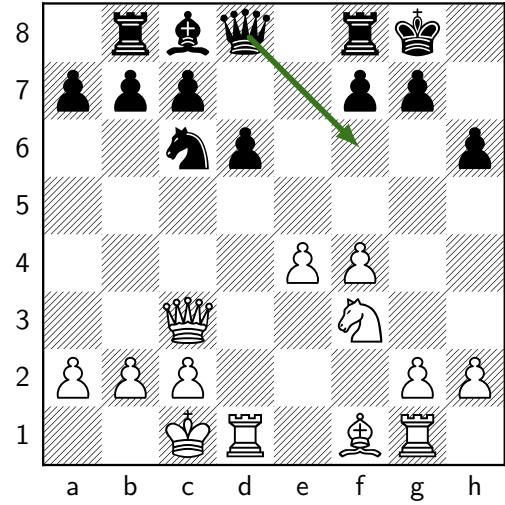
(a)



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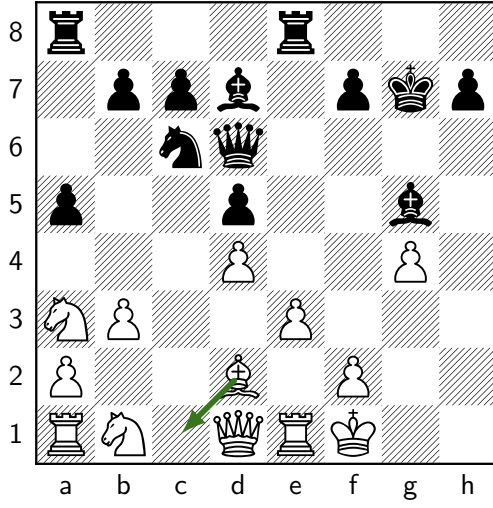


(c)

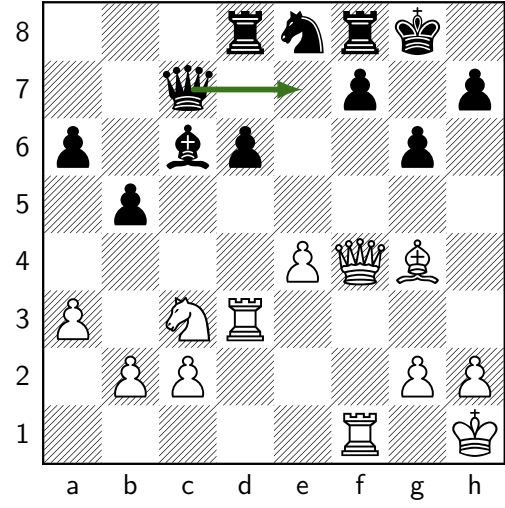


(d)

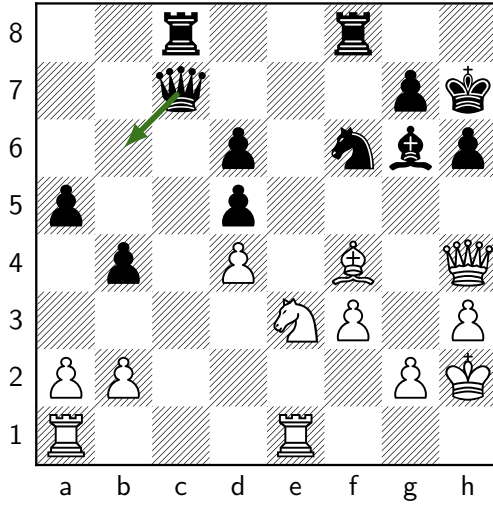
Figure 13: **Move 19, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#451` (ply 39) [structural distance $d=74$, QAP obj=93]: the master played Rd8; under μ our Bc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ne8, our Re1 as their Qc7; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1682` (ply 47) [structural distance $d=91$, QAP obj=45]: the master played Nce7; under μ our Bc1 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra8, our Pf2 as their Qe8; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#913` (ply 23) [structural distance $d=91$, QAP obj=43]: the master played Qf6; under μ our Bc1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Rb8, our Pg4 as their Bc8; transferred move c1d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd2 (summed QAP objective 224, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



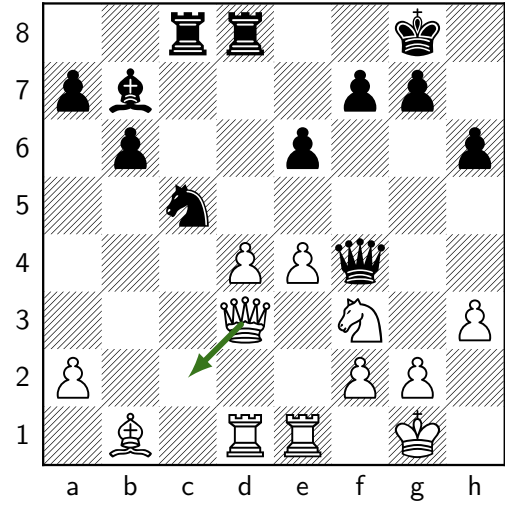
(a)



(b)

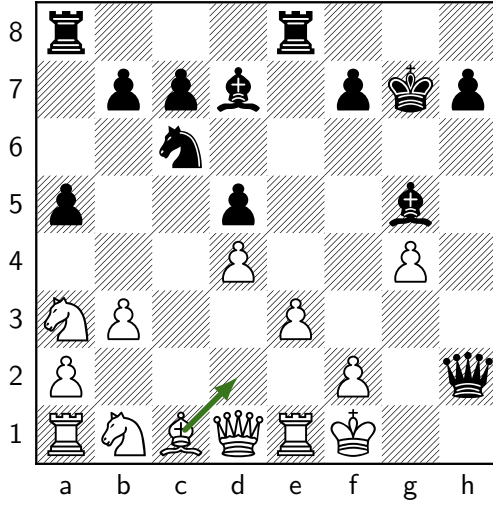


(c)

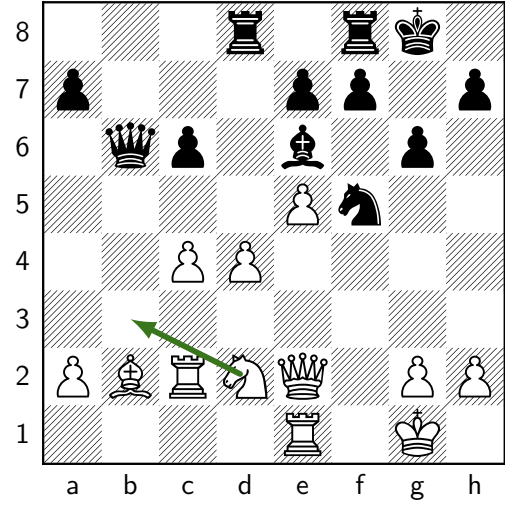


(d)

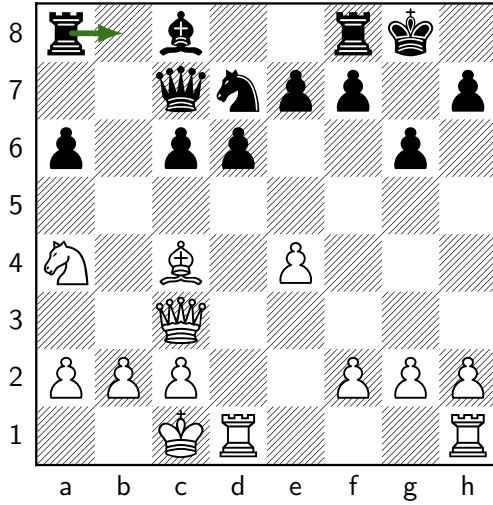
Figure 14: **Move 20, White: Bc1.** (a) Our position; Melanie plays Bc1 (d2c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#451` (ply 41) [structural distance $d=71$, QAP obj=112]: the master played Qe7; under μ our Bd2 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ne8, our Bd2 as their Rf8; transferred move d2c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2304` (ply 51) [structural distance $d=87$, QAP obj=74]: the master played Qb6; under μ our Bd2 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Rc8, our Pe3 as their Nf6; transferred move d2c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#949` (ply 36) [structural distance $d=86$, QAP obj=42]: the master played Qc2; under μ our Bd2 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Bb1, our Qd1 as their Rd1; transferred move d2c1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc1 (summed QAP objective 256, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



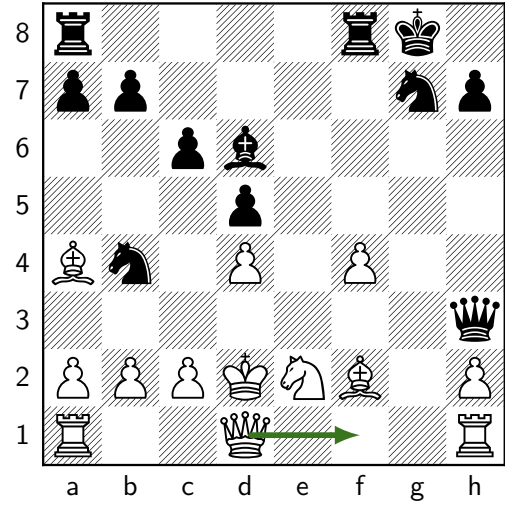
(a)



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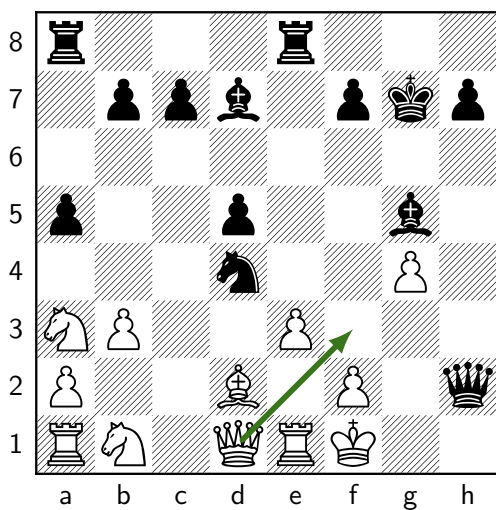


(c)

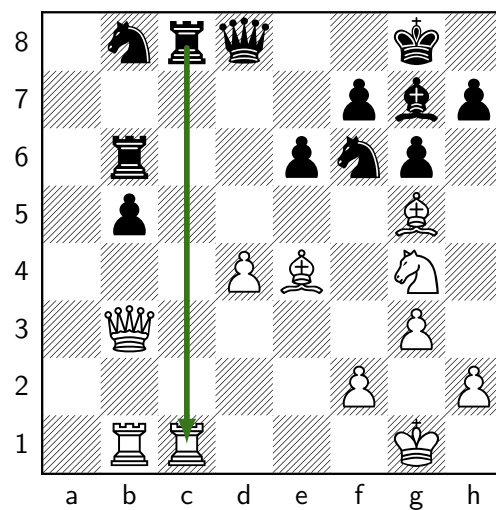


(d)

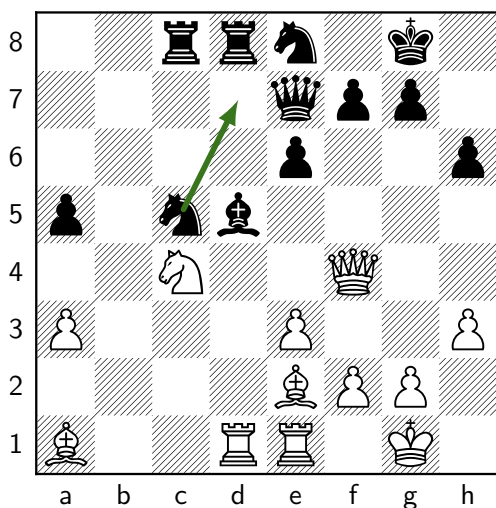
Figure 15: **Move 21, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1936` (ply 40) [structural distance $d=95$, QAP obj=84]: the master played Nb3; under μ our Bc1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pe3 as their Bb2, our Qd1 as their Rc2; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2532` (ply 29) [structural distance $d=83$, QAP obj=72]: the master played Rb8; under μ our Bc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Bc8, our Qd1 as their Rf8; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2768` (ply 36) [structural distance $d=97$, QAP obj=53]: the master played Qf1; under μ our Bc1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra1, our Ra1 as their Pb2; transferred move c1d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd2 (summed QAP objective 232, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



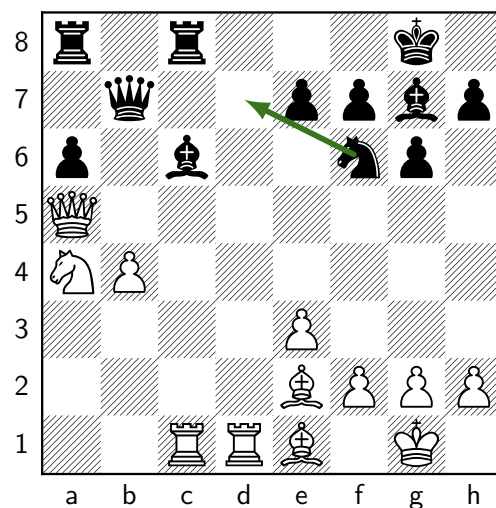
(a)



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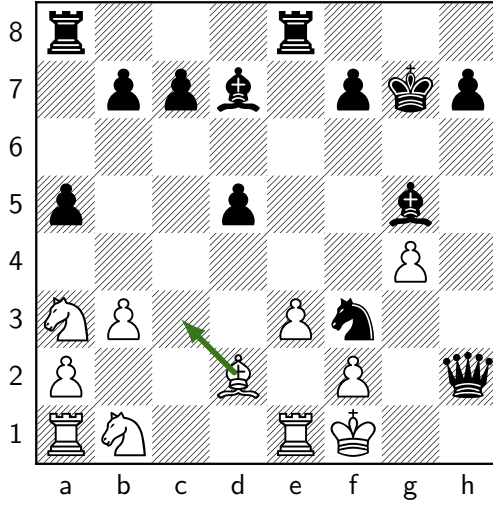


(c)

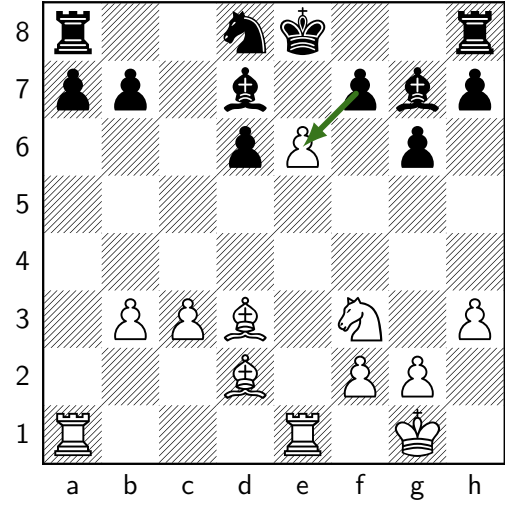


(d)

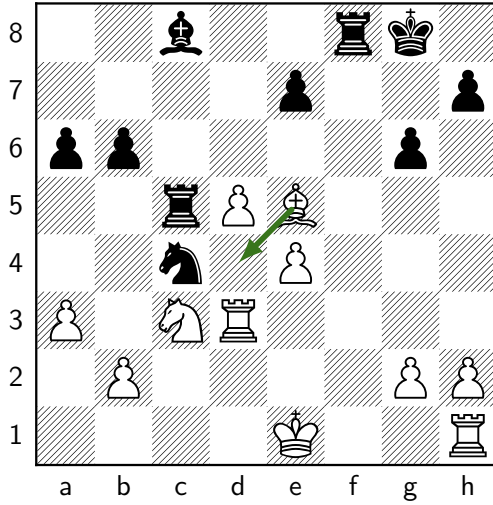
Figure 16: **Move 22, White: Qf3.** (a) Our position; Melanie plays Qf3 (d1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1795` (ply 47) [structural distance $d=103$, QAP obj=40]: the master played Rxc1+; under μ our Qd1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pg4 as their Nb8, our Bd2 as their Qd8; transferred move d1f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1609` (ply 45) [structural distance $d=107$, QAP obj=91]: the master played Nd7; under μ our Nb1 plays the role of that game’s Nc4; cross-type roles: our Bd2 as their Rd8, our Qd1 as their Ne8; transferred move b1c3. (d) Retrieved from `ccr1_top_p0000#3819` (ply 43) [structural distance $d=98$, QAP obj=78]: the master played Nd7; under μ our Na3 plays the role of that game’s Nf3; cross-type roles: our pb7 as their Ra8, our Bd2 as their Rc8; transferred move a3c2. Synthesis: 1 retrieved analog transferred to Qf3 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



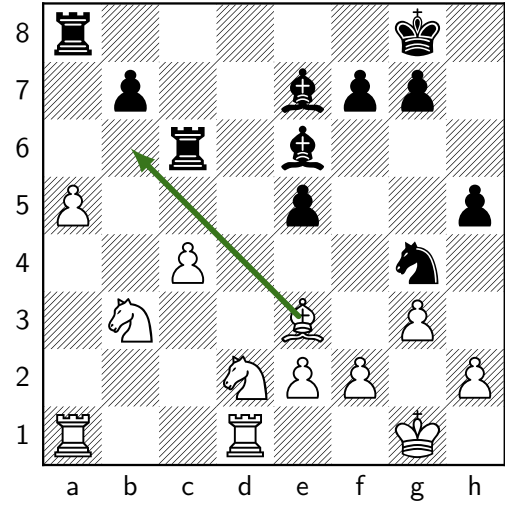
(a)



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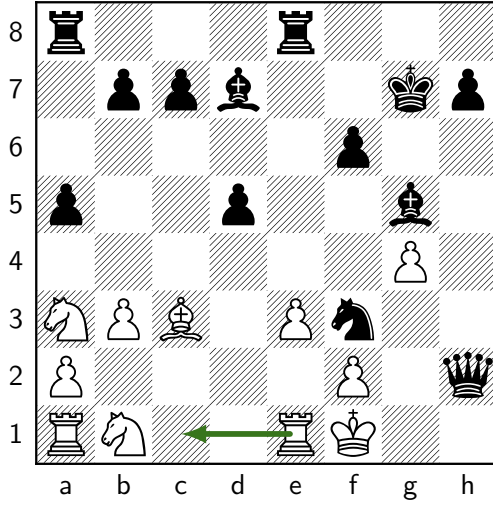


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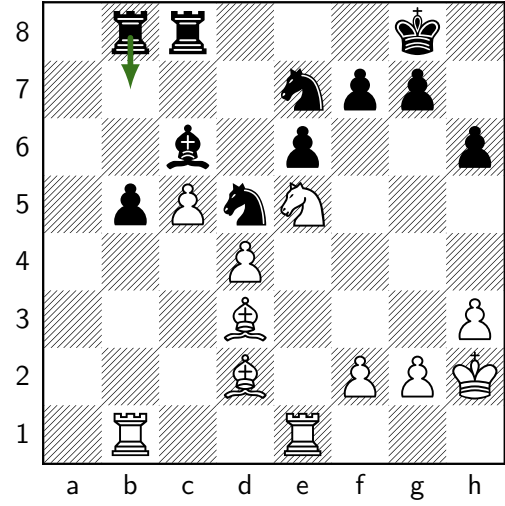


(d)

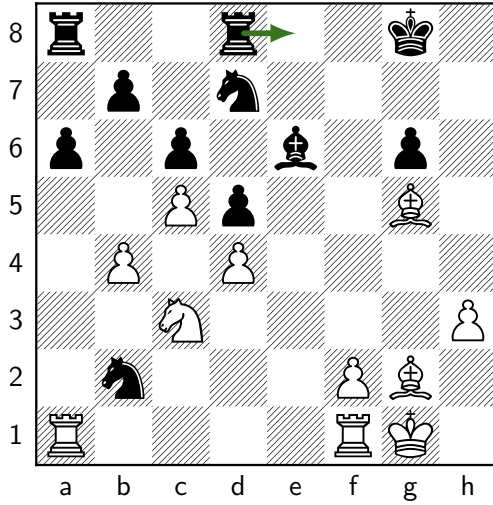
Figure 17: **Move 23, White: Bc3+.** (a) Our position; Melanie plays Bc3+ (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#234` (ply 31) [structural distance $d=115$, QAP obj=52]: the master played **fxe6**; under μ our *Bd2* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Ra8*, our *pa5* as their *Nd8*; transferred move **d2c3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#171` (ply 44) [structural distance $d=120$, QAP obj=35]: the master played **Bd4**; under μ our *Bd2* plays the role of that game’s *Be5*; cross-type roles: our *bd7* as their *Ph2*, our *Na3* as their *Pa3*; transferred move **d2c3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#685` (ply 40) [structural distance $d=101$, QAP obj=54]: the master played **Bb6**; under μ our *Bd2* plays the role of that game’s *Be3*; cross-type roles: our *ph7* as their *Ra1*, our *bd7* as their *Rd1*; transferred move **d2a5**. Synthesis: 2 retrieved analogs transferred to Bc3+ (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



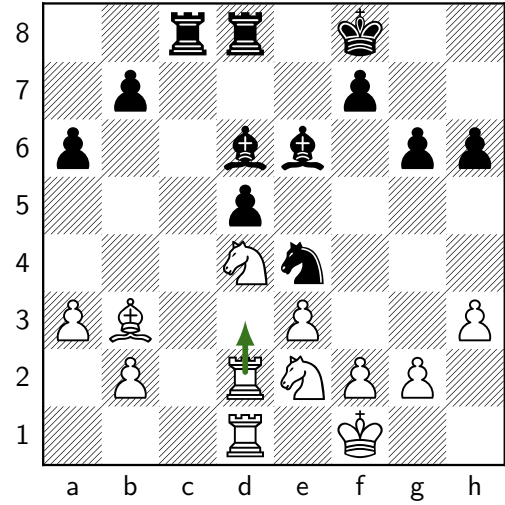
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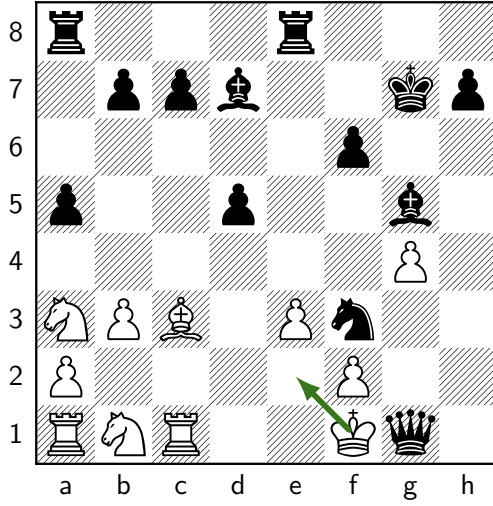


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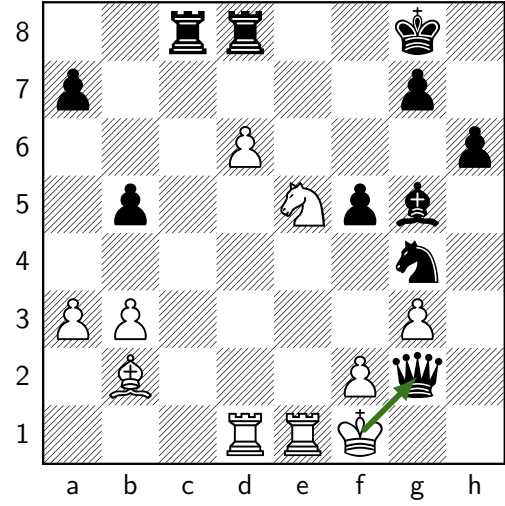


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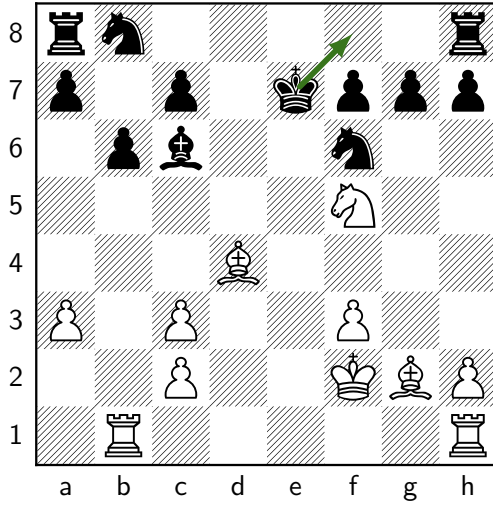
Figure 18: **Move 24, White: Rc1.** (a) Our position; Melanie plays Rc1 (e1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#701` (ply 63) [structural distance $d=102$, QAP obj=39]: the master played Rb7; under μ our *Re1* plays the role of that game’s *Rb1*; cross-type roles: our *Bc3* as their *Rc8*, our *Pf2* as their *Ne7*; transferred move e1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3116` (ply 45) [structural distance $d=112$, QAP obj=38]: the master played Re8; under μ our *Re1* plays the role of that game’s *Rd1*; cross-type roles: our *Bc3* as their *Nd7*, our *Pg4* as their *Be6*; transferred move e1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1936` (ply 58) [structural distance $d=100$, QAP obj=38]: the master played Rd3; under μ our *Re1* plays the role of that game’s *Rd2*; cross-type roles: our *ph7* as their *Rd1*, our *Pg4* as their *Ne2*; transferred move e1c1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rc1 (summed QAP objective 173, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



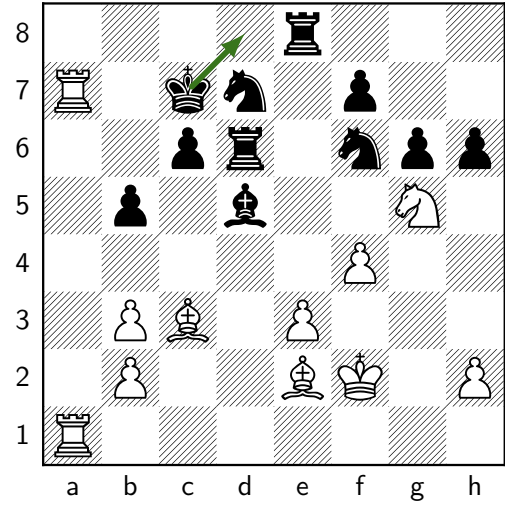
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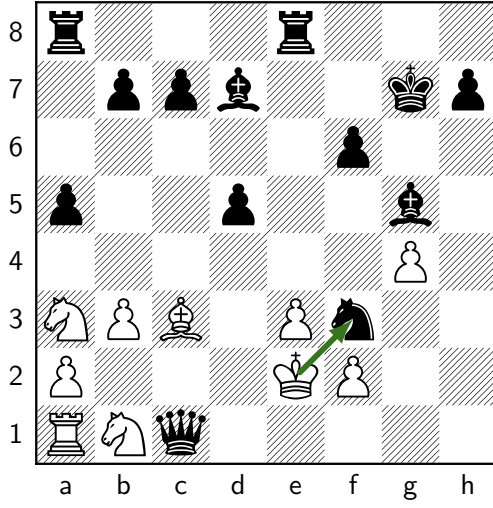


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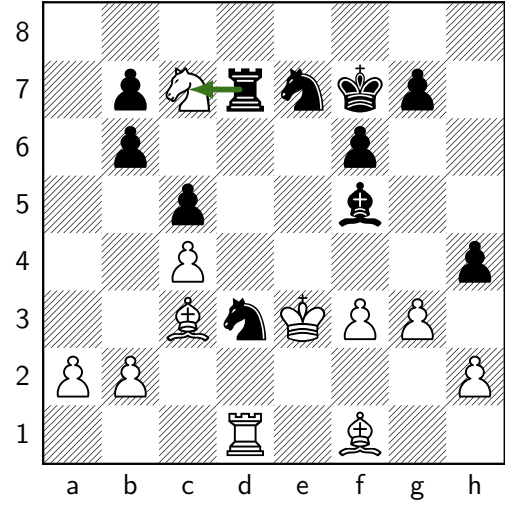


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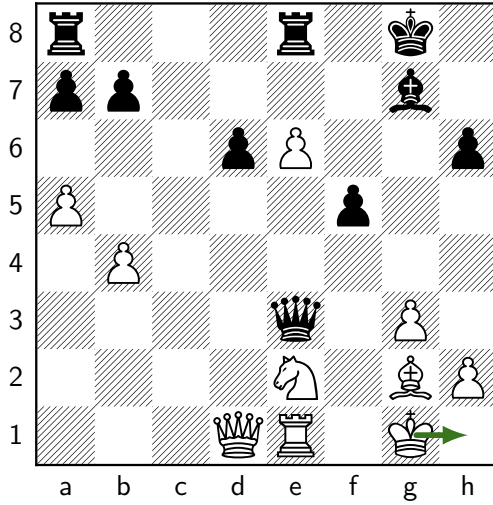
Figure 19: **Move 25, White: Ke2.** (a) Our position; Melanie plays Ke2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2122` (ply 60) [structural distance $d=117$, QAP obj=92]: the master played Kxg2; under μ our *Kf1* plays the role of that game’s *Kf1*; cross-type roles: our *pf6* as their *Bb2*, our *Ra1* as their *Pa3*; transferred move f1e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#555` (ply 33) [structural distance $d=98$, QAP obj=61]: the master played Kf8; under μ our *Kf1* plays the role of that game’s *Ke2*; cross-type roles: our *Rc1* as their *Nb8*, our *bd7* as their *Rh8*; transferred move f1e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#487` (ply 49) [structural distance $d=115$, QAP obj=48]: the master played Kd8; under μ our *Kf1* plays the role of that game’s *Kc2*; cross-type roles: our *Bc3* as their *Re8*, our *qg1* as their *ra7*; transferred move f1e2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ke2 (summed QAP objective 284, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



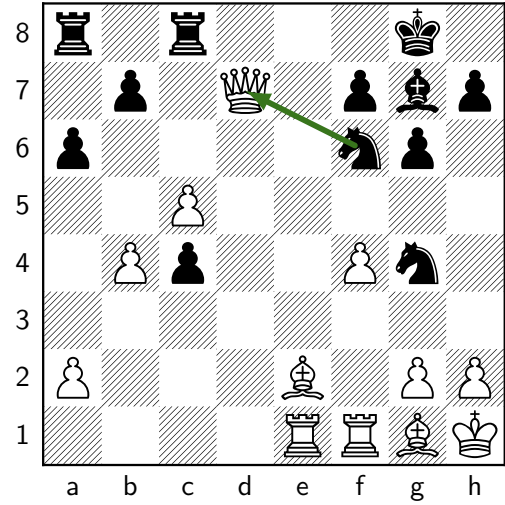
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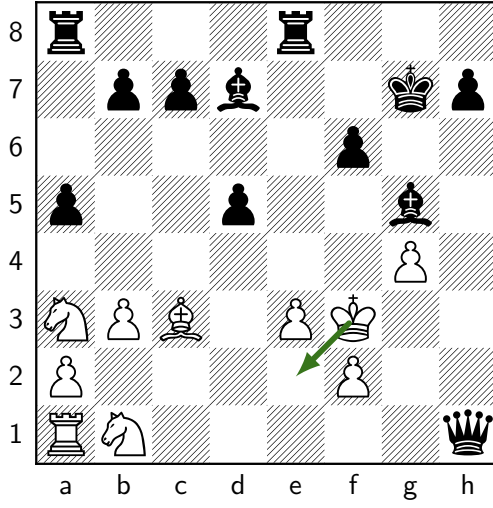


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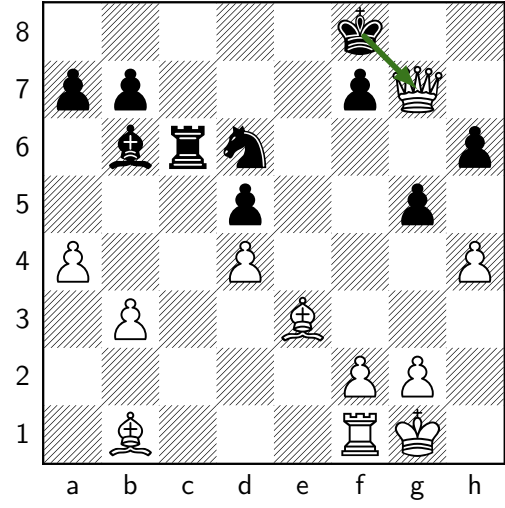


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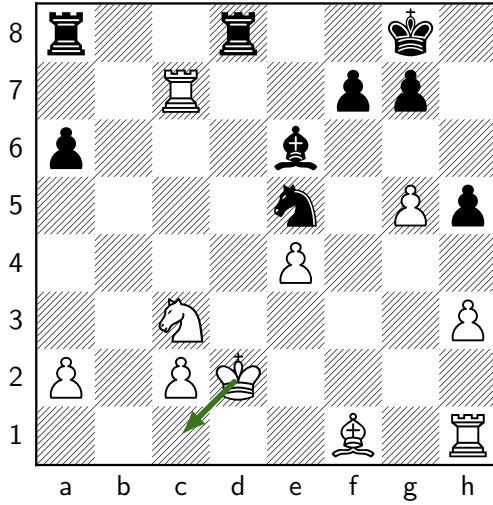
Figure 20: **Move 26, White: Kxf3**. (a) Our position; Melanie plays Kxf3 (e2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3121` (ply 59) [structural distance $d=119$, QAP obj=51]: the master played Rxc7; under μ our Ke2 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ne7, our bd7 as their Pc5; transferred move e2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1308` (ply 60) [structural distance $d=118$, QAP obj=37]: the master played Kh1; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our Na3 as their Qd1, our Pg4 as their Bg2; transferred move e2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3502` (ply 51) [structural distance $d=102$, QAP obj=21]: the master played Nxd7; under μ our Ke2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our bd7 as their Ra8, our Bc3 as their Rc8; transferred move e2f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kxf3 (summed QAP objective 109, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



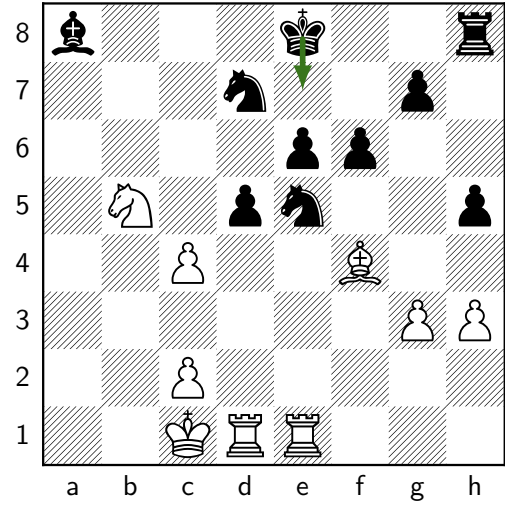
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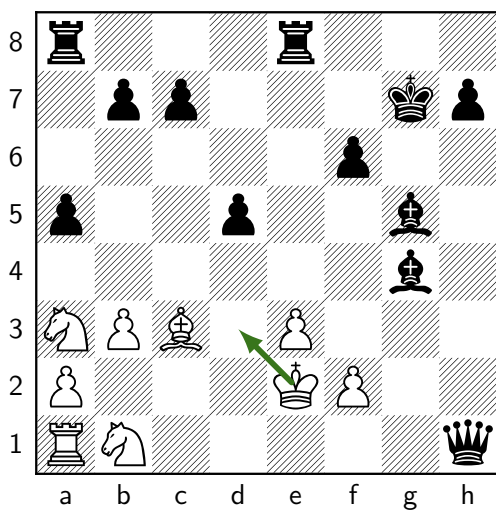


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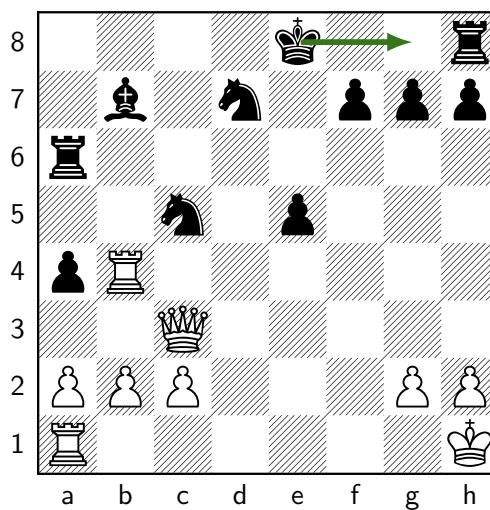


(d)

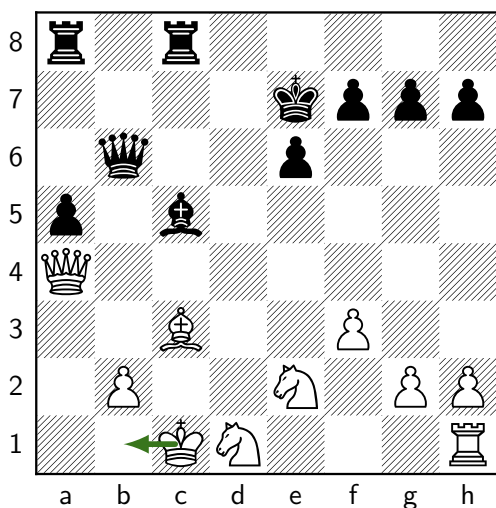
Figure 21: **Move 27, White: Ke2.** (a) Our position; Melanie plays Ke2 (f3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2552` (ply 61) [structural distance $d=89$, QAP obj=56]: the master played Kxg7; under μ our Kf3 plays the role of that game’s Kf1; cross-type roles: our qh1 as their Bb6, our Na3 as their Rc6; transferred move f3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1865` (ply 40) [structural distance $d=88$, QAP obj=29]: the master played Kc1; under μ our Kf3 plays the role of that game’s Kd2; cross-type roles: our pf6 as their Bf1, our re8 as their Nc3; transferred move f3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1640` (ply 53) [structural distance $d=91$, QAP obj=16]: the master played Ke7; under μ our Kf3 plays the role of that game’s Ke1; cross-type roles: our pd5 as their Nd7, our bd7 as their Pf6; transferred move f3e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke2 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



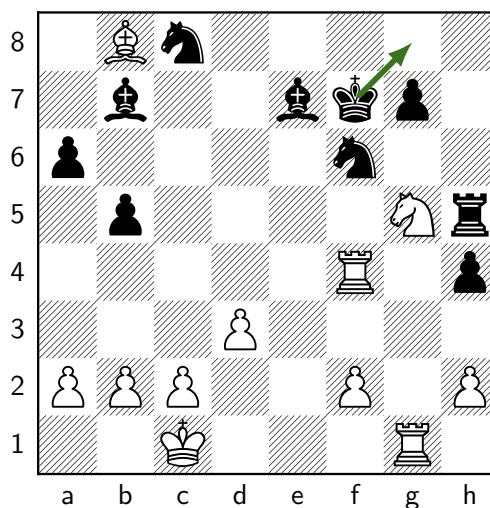
(a)



(b)

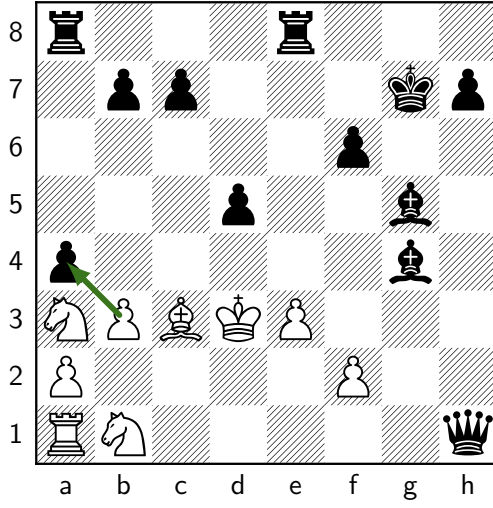


(c)

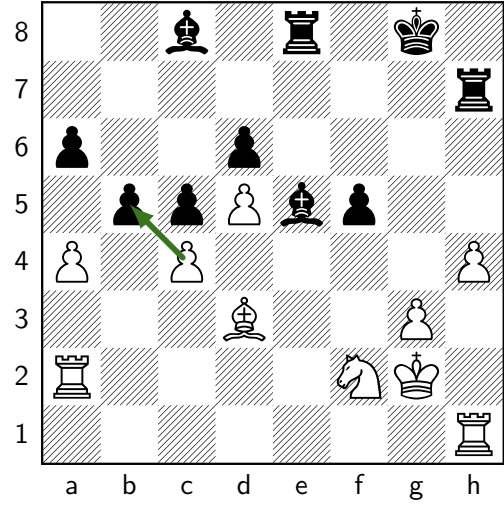


(d)

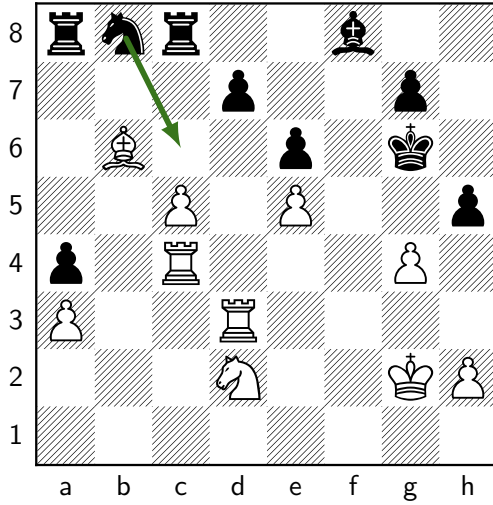
Figure 22: **Move 28, White: Kd3.** (a) Our position; Melanie plays Kd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3622` (ply 43) [structural distance $d=91$, QAP obj=16]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Rh8*, our *Pf2* as their *Nd7*; transferred move e2d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2585` (ply 42) [structural distance $d=82$, QAP obj=13]: the master played Kb1; under μ our *Ke2* plays the role of that game’s *Kc1*; cross-type roles: our *ph7* as their *Nd1*, our *Pf2* as their *Ne2*; transferred move e2d3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3256` (ply 61) [structural distance $d=94$, QAP obj=76]: the master played Kg8; under μ our *Ke2* plays the role of that game’s *Kf2*; cross-type roles: our *Pa2* as their *bb8*, our *ra8* as their *Bb7*; transferred move e2d2. Synthesis: 3 retrieved analogs transferred to Kd3 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



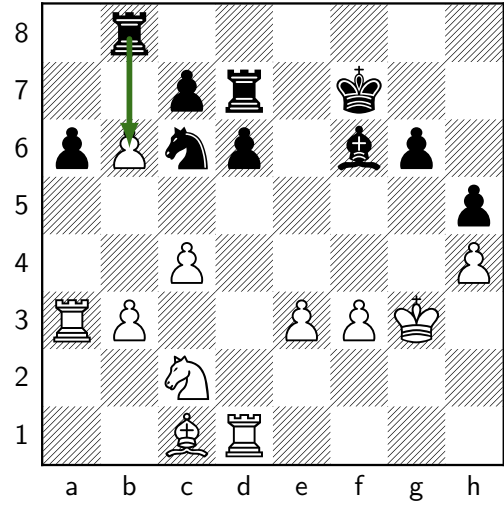
(a)



(b)

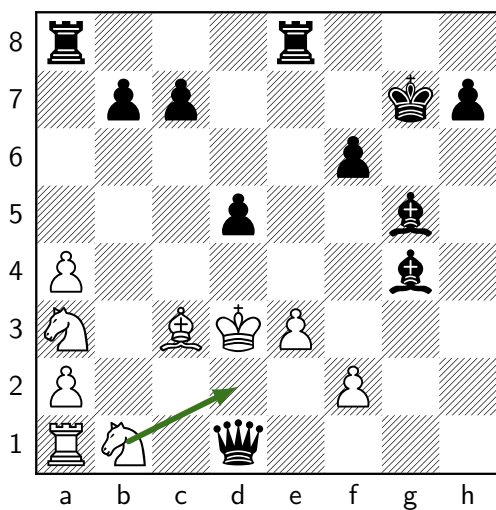


(c)

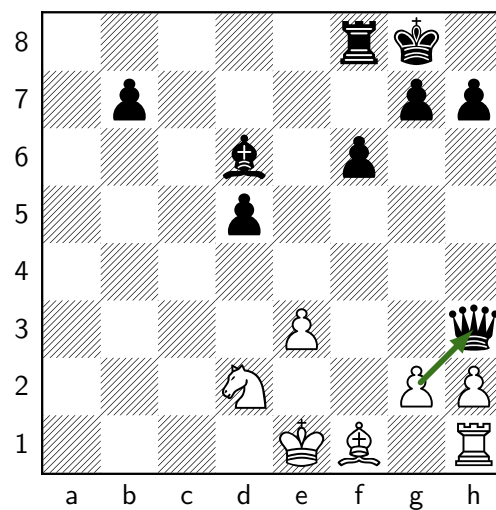


(d)

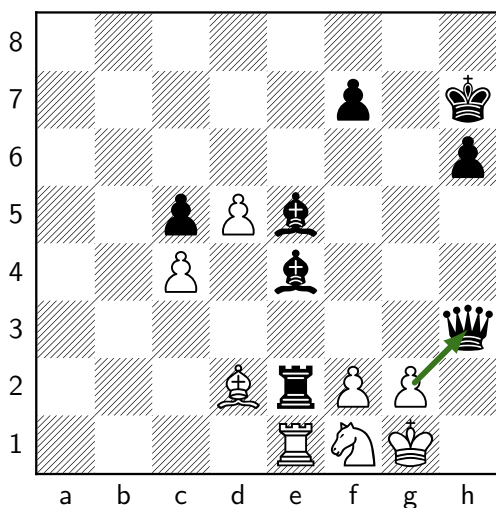
Figure 23: **Move 29, White: bxa4.** (a) Our position; Melanie plays bxa4 (b3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#357` (ply 66) [structural distance $d=97$, QAP obj=128]: the master played `cxb5`; under μ our `Pb3` plays the role of that game’s `Pc4`; cross-type roles: our `Pa2` as their `Ra2`, our `pb7` as their `Nf2`; transferred move `b3a4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1871` (ply 65) [structural distance $d=97$, QAP obj=75]: the master played `Nc6`; under μ our `Na3` plays the role of that game’s `Nb1`; cross-type roles: our `bg4` as their `Ra8`, our `qh1` as their `Rc8`; transferred move `a3b5`. (d) Retrieved from `ccr1_top_p0000#1818` (ply 69) [structural distance $d=96$, QAP obj=66]: the master played `Rxb6`; under μ our `Pb3` plays the role of that game’s `Rb1` (a **cross-type** correspondence); cross-type roles: our `Na3` as their `Rd7`, our `Pa2` as their `Nc6`; transferred move `b3b4`. Synthesis: 1 retrieved analog transferred to `bxa4` (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



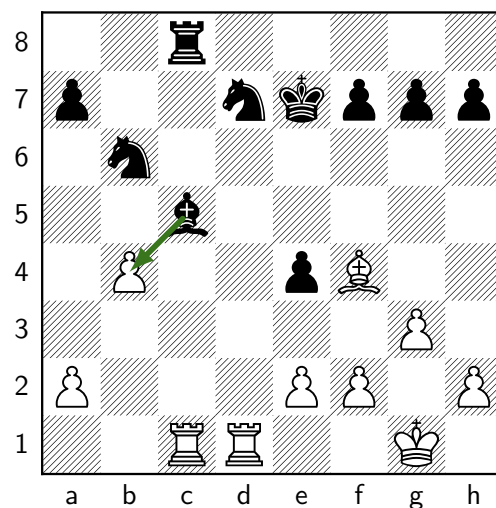
(a)



(b)

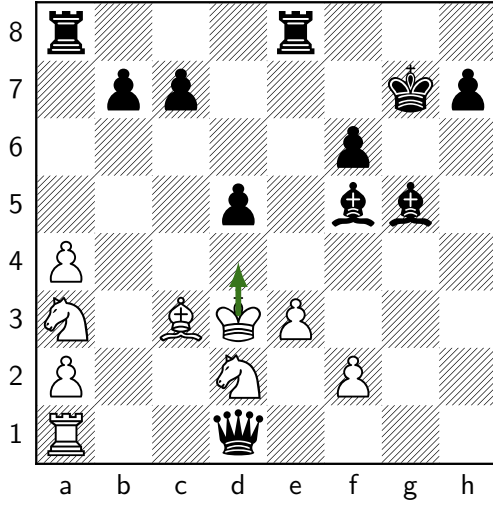


(c)

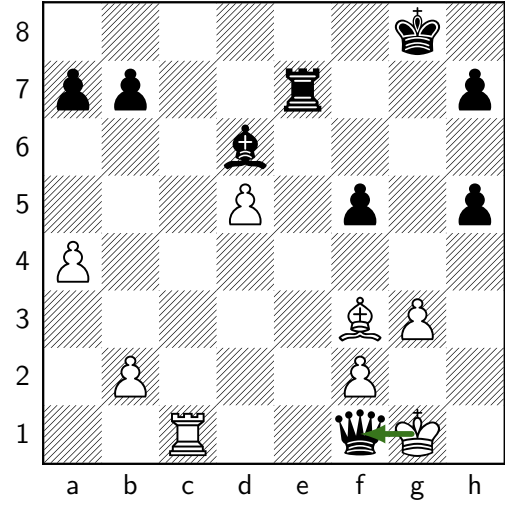


(d)

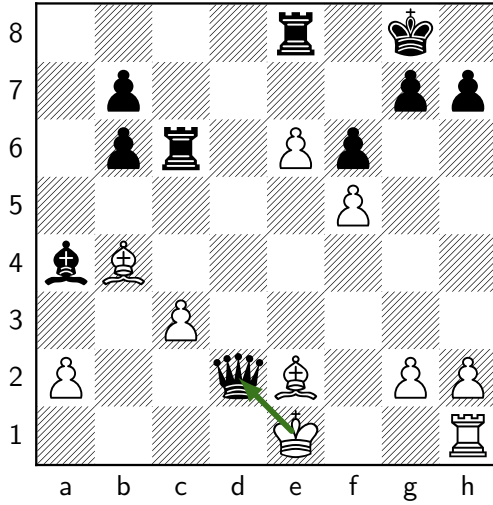
Figure 24: **Move 30, White: Nd2.** (a) Our position; Melanie plays Nd2 (b1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#3639` (ply 48) [structural distance $d=107$, QAP obj=30]: the master played `gxh3`; under μ our `Nb1` plays the role of that game’s `Pg2` (a **cross-type** correspondence); cross-type roles: our `Na3` as their `Bf1`, our `Pe3` as their `Nd2`; transferred move `b1d2` (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#1220` (ply 74) [structural distance $d=99$, QAP obj=17]: the master played `gxh3`; under μ our `Nb1` plays the role of that game’s `Pg2` (a **cross-type** correspondence); cross-type roles: our `pf6` as their `Re1`, our `Ra1` as their `Nf1`; transferred move `b1d2` (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#872` (ply 45) [structural distance $d=102$, QAP obj=13]: the master played `Bxb4`; under μ our `Nb1` plays the role of that game’s `Bc4` (a **cross-type** correspondence); cross-type roles: our `Bc3` as their `Pg7`, our `bg4` as their `Ph7`; transferred move `b1d2` (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nd2 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



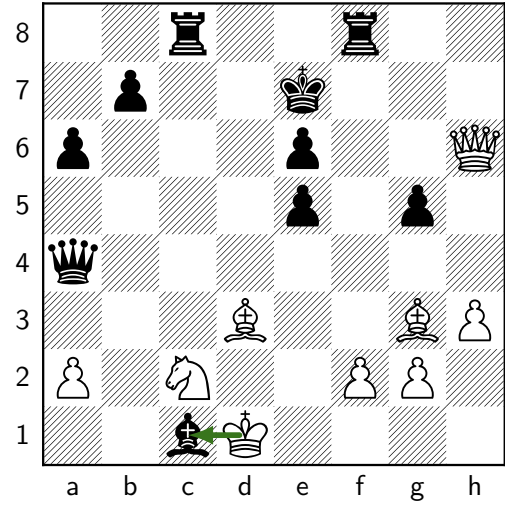
(a)



(b)

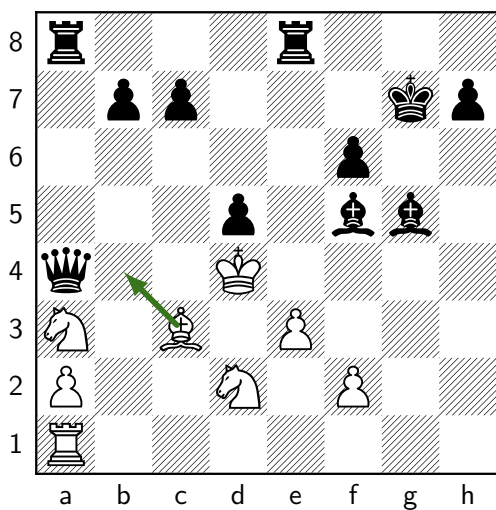


(c)

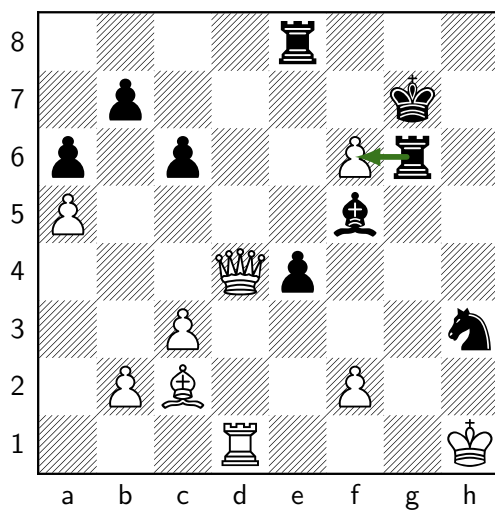


(d)

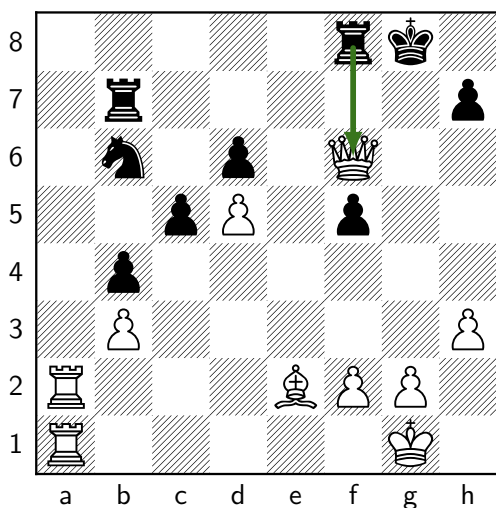
Figure 25: **Move 31, White: Kd4.** (a) Our position; Melanie plays Kd4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#332` (ply 60) [structural distance $d=91$, QAP obj=43]: the master played Kxf1; under μ our *Kd3* plays the role of that game’s *Kg1*; cross-type roles: our *bf5* as their *qf1*, our *Nd2* as their *Pf2*; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#16` (ply 56) [structural distance $d=89$, QAP obj=36]: the master played Kxd2; under μ our *Kd3* plays the role of that game’s *Ke1*; cross-type roles: our *Pe3* as their *Rh1*, our *Ra1* as their *Pa2*; transferred move d3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1172` (ply 50) [structural distance $d=98$, QAP obj=23]: the master played Kxc1; under μ our *Kd3* plays the role of that game’s *Kd1*; cross-type roles: our *Ra1* as their *Pa2*, our *Na3* as their *Ph3*; transferred move d3d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd4 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



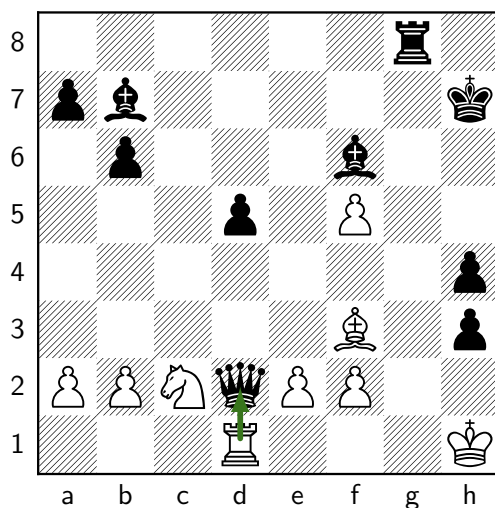
(a)



(b)

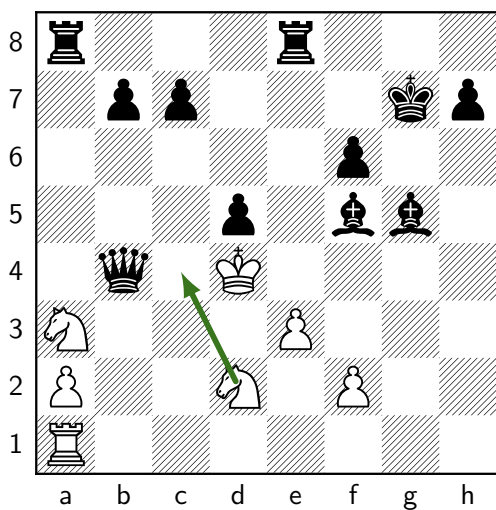


(c)

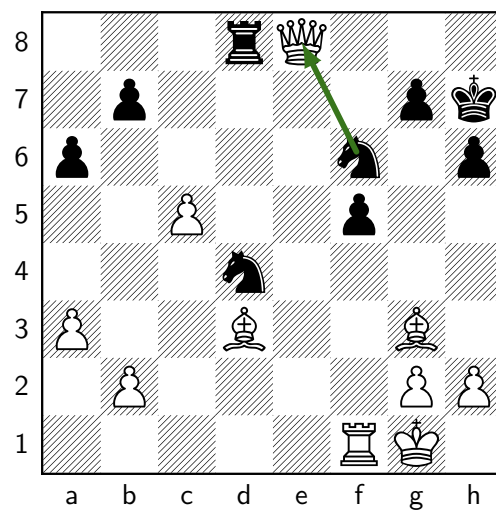


(d)

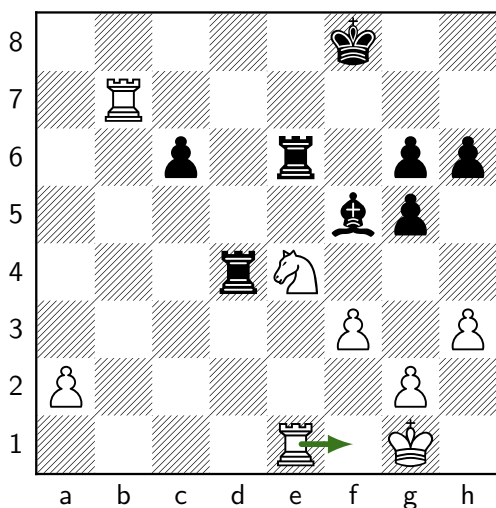
Figure 26: **Move 32, White: Bb4**. (a) Our position; Melanie plays Bb4 (c3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1206` (ply 67) [structural distance $d=88$, QAP obj=57]: the master played Rxf6; under μ our Bc3 plays the role of that game’s Rg3 (a **cross-type** correspondence); cross-type roles: our Na3 as their Pa6, our qa4 as their pf6; transferred move c3b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1428` (ply 53) [structural distance $d=102$, QAP obj=26]: the master played Rxf6; under μ our Bc3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Rb7, our bg5 as their qf6; transferred move c3b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2389` (ply 62) [structural distance $d=97$, QAP obj=26]: the master played Rxd2; under μ our Bc3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our bf5 as their Nc2, our Pf2 as their qd2; transferred move c3b4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bb4 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



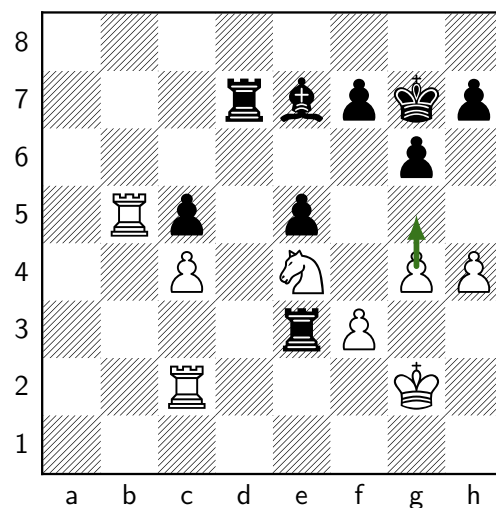
(a)



(b)

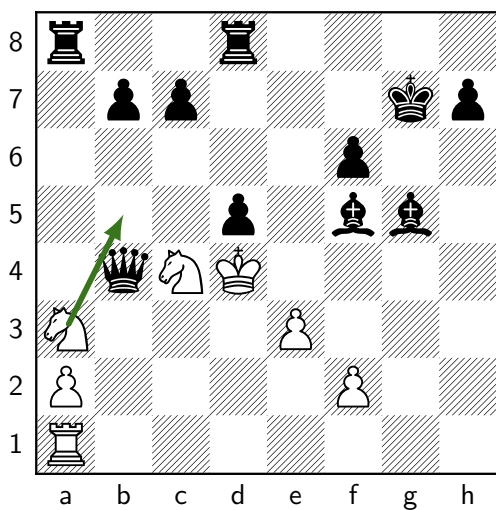


(c)

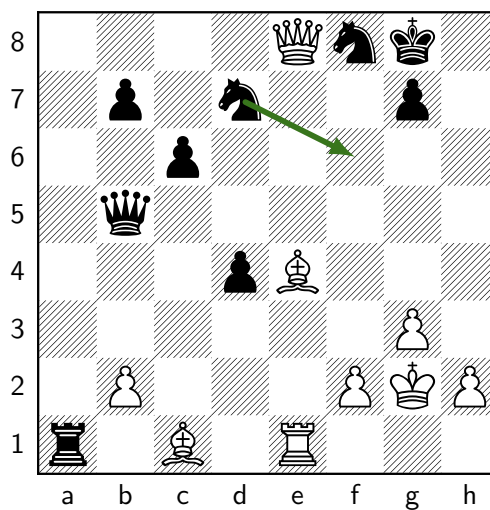


(d)

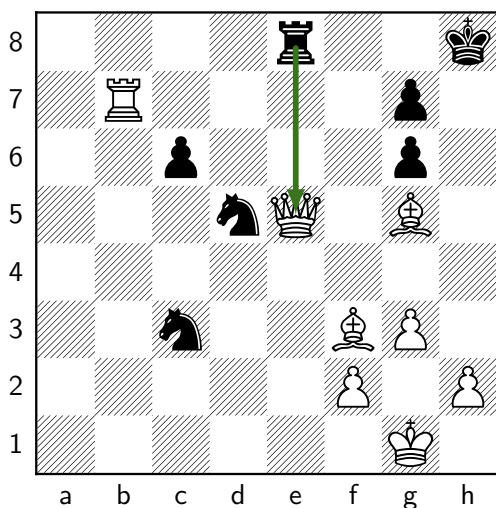
Figure 27: **Move 33, White: Ndc4**. (a) Our position; Melanie plays Ndc4 (d2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2274` (ply 79) [structural distance $d=86$, QAP obj=30]: the master played Nxe8; under μ our Nd2 plays the role of that game’s Nf3; cross-type roles: our Na3 as their Rd8, our pd5 as their qe8; transferred move d2c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2233` (ply 66) [structural distance $d=88$, QAP obj=23]: the master played Rf1; under μ our Nd2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Na3 as their Ph3, our Pe3 as their Ne4; transferred move d2c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#221` (ply 80) [structural distance $d=88$, QAP obj=18]: the master played g5; under μ our Nd2 plays the role of that game’s Pg4 (a **cross-type** correspondence); cross-type roles: our Pe3 as their Rc2, our qb4 as their re3; transferred move d2c4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ndc4 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



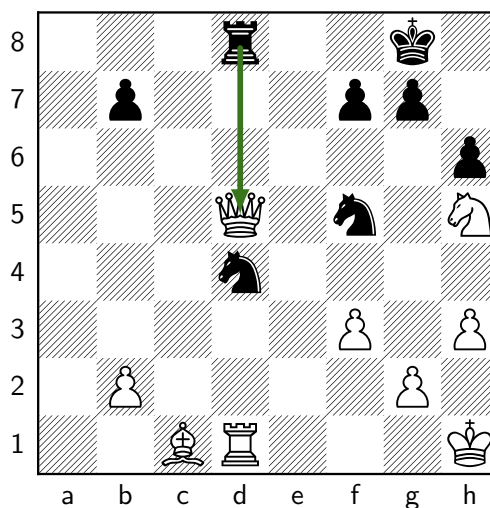
(a)



(b)



(c)



(d)

Figure 28: **Move 34, White: Nb5**. (a) Our position; Melanie plays Nb5 (a3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1638` (ply 61) [structural distance $d=91$, QAP obj=103]: the master played Nf6; under μ our Na3 plays the role of that game’s Nd2; cross-type roles: our Ra1 as their Qb5, our pb7 as their be4; transferred move a3b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1018` (ply 69) [structural distance $d=107$, QAP obj=58]: the master played Rxe5; under μ our Na3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our pd5 as their rb7, our Nc4 as their Pg7; transferred move a3b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#875` (ply 53) [structural distance $d=108$, QAP obj=54]: the master played Rxd5; under μ our Na3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Pf7, our bg5 as their nh5; transferred move a3b5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nb5 (summed QAP objective 340, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.